Diary Data Analysis

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2025-08-12

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0) Tables

```
## `summarise()` has grouped output by 'Training_Condition'. You can override
## using the `.groups` argument.
## `summarise()` has grouped output by 'Training_Condition'. You can override
## using the `.groups` argument.
## `summarise()` has grouped output by 'Training_Condition'. You can override
## using the `.groups` argument.
## `summarise()` has grouped output by 'Training_Condition'. You can override
## using the `.groups` argument.
```

Table 1 : child ASB

Training_Condition	diaryTP	n
CONTROL	2	40
CONTROL	3	40
TRAIN	2	34
TRAIN	3	34

```
## [1] "n participants total = "
```

```
## [1] 74
```

Table 2: child PSB

Training_Condition	diaryTP	n
CONTROL	2	40
CONTROL	3	40
TRAIN	2	34
TRAIN	3	34

```
## [1] "n participants total = "
```

```
## [1] 74
```

Table 2: teach ASB

Training_Condition	diaryTP	n
CONTROL	2	51
CONTROL	3	51
TRAIN	2	44
TRAIN	3	44

```
## [1] "n participants total = "
```

```
## [1] 95
```

Table 4: teach PSB

Training_Condition	diaryTP	n
CONTROL	2	51
CONTROL	3	51
TRAIN	2	44
TRAIN	3	44

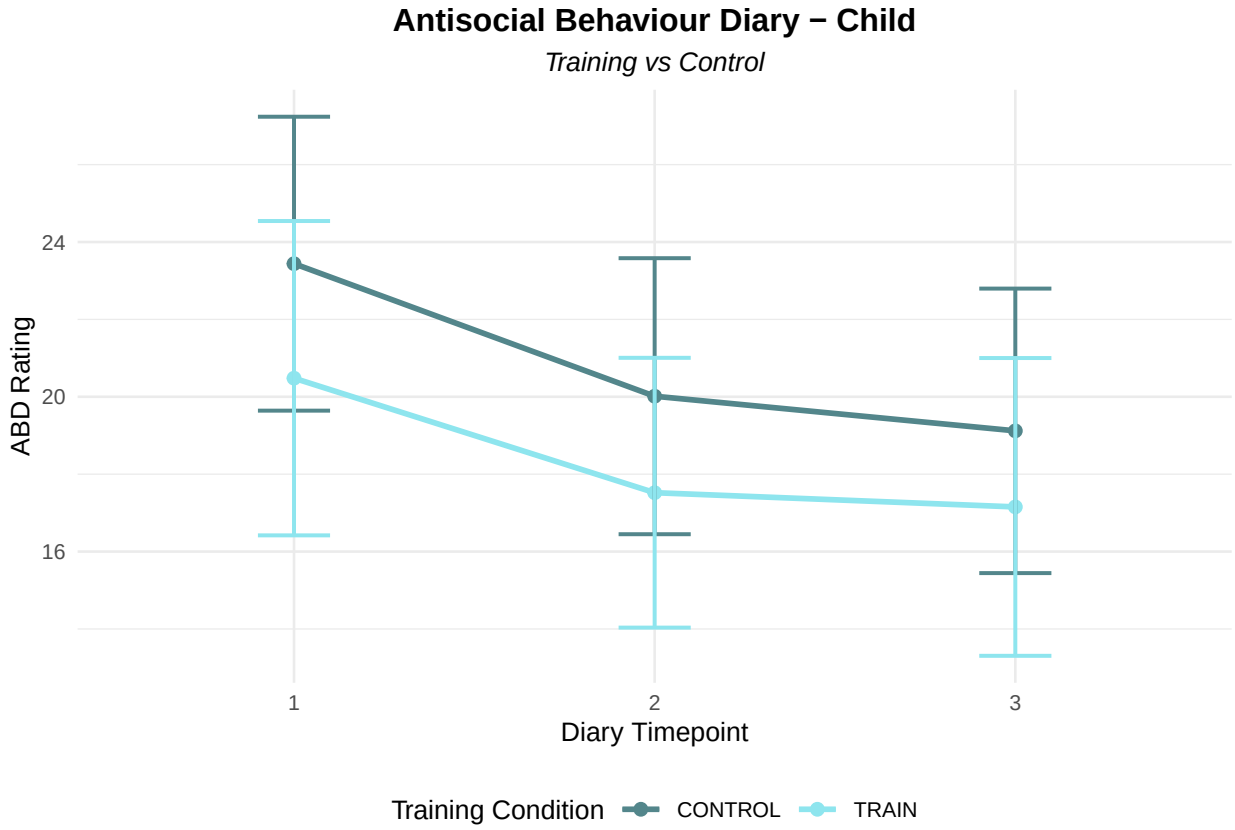
```
## [1] "n participants total = "
```

```
## [1] 95
```

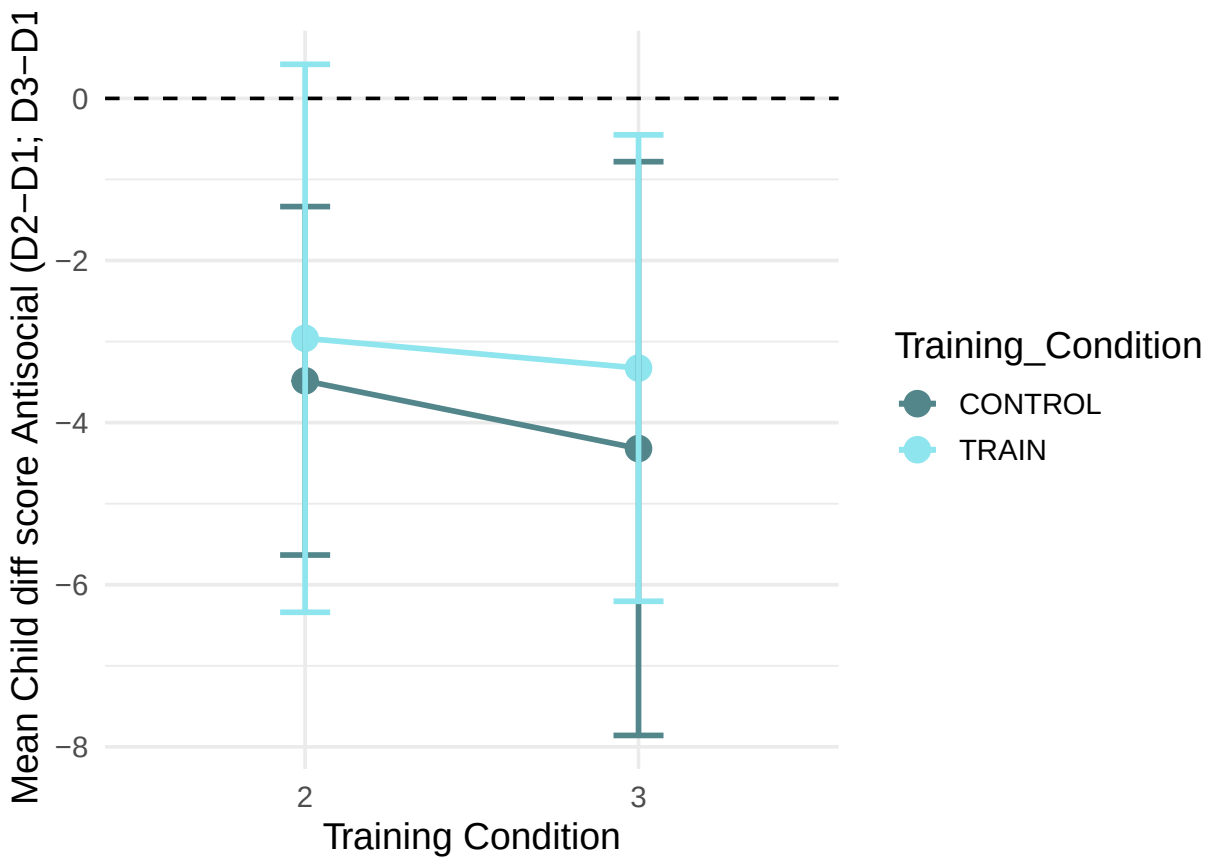
2) Impact of Training on Child Rated Antisocial Behaviour at Two Months (T3)

2.1) Plotting Data - Impact of training over time on child ratings of antisocial behaviour

First plotting data:



Difference Scores:



2.2) Preregistered LMM - Random slopes always throws error. From here, not including in any model.

Trying prespecified model: this throws the following error: 'Error: number of observations (=192) <= number of random effects (=192) for term (1 + diaryTP | ID_numeric); the random-effects parameters and the residual variance (or scale parameter) are probably unidentifiable'

From my understanding this error comes from us not having enough data per ID at each timepoint to estimate random slopes. We only have two datapoints per ID (D2 and D3), so there is no within participant variance.

So, removing random slope.

```
# model that gives error:
#
# model_c_time_asb <- lmer(A_sum_child ~ ABD_sum_D1 + diaryTP +
#                            Training_Condition + diaryTP*Training_Condition +
#                          (1|ID_numeric + diaryTP|ID_numeric),
#                          data =  child_abd_d2d3_longformat)



model_c_time_asb <- lmer(A_sum_child ~ ABD_sum_D1 + diaryTP +
                           Training_Condition + diaryTP*Training_Condition +
               (1 | ID_numeric),
                      data =  child_abd_d2d3_longformat)

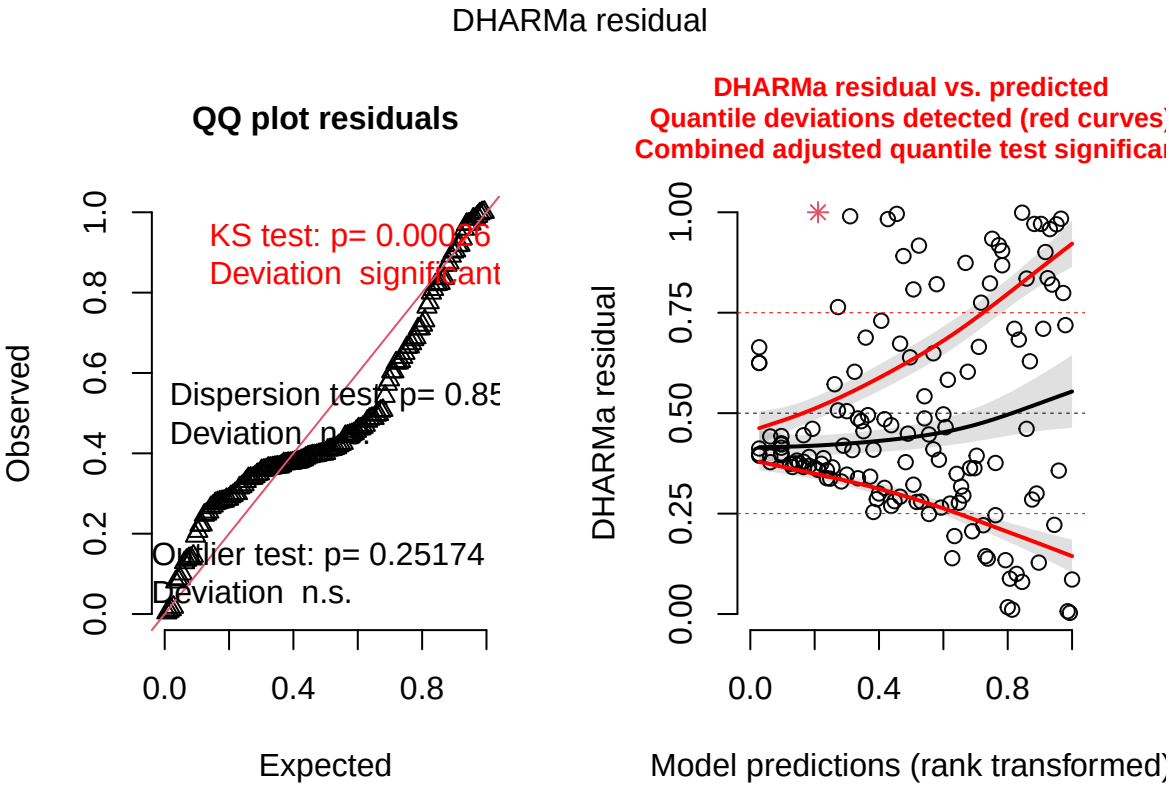


summary(model_c_time_asb)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_sum_child ~ ABD_sum_D1 + diaryTP + Training_Condition + diaryTP *
##     Training_Condition + (1 | ID_numeric)
##    Data: child_abd_d2d3_longformat
##
## REML criterion at convergence: 1185.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9381 -0.3098 -0.1349  0.2649  4.2307
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ID_numeric (Intercept) 129.1    11.36
##  Residual               136.6    11.69
## Number of obs: 145, groups:  ID_numeric, 73
##
## Fixed effects:
##                                 Estimate Std. Error        df t value
## (Intercept)                      5.38650    3.09907  99.08050   1.738
## ABD_sum_D1                       0.63975    0.06988  71.08882   9.155
## diaryTP3                        -1.08389    2.63872  70.78582  -0.411
```

```
## Training_ConditionTRAIN           -0.96815    3.84438 112.62228  -0.252
## diaryTP3:Training_ConditionTRAIN   0.71662    3.90420  70.42659   0.184
##                                  Pr(>|t|)
## (Intercept)                        0.0853 .
## ABD_sum_D1                        1.2e-13 ***
## diaryTP3                           0.6825
## Training_ConditionTRAIN            0.8016
## diaryTP3:Training_ConditionTRAIN   0.8549
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) ABD__D dryTP3 T_CTRA
## ABD_sum_D1  -0.555
## diaryTP3    -0.427  0.016
## Trnn_CTRAIN -0.599  0.075  0.338
## dTP3:T_CTRA  0.288 -0.011 -0.676 -0.504
```

2.1.1) Assumption Check:



2.1.2) Complexity check: Including an interaction does not improve model fit.

```
model_c_time_simple <- lmer(A_sum_child ~ ABD_sum_D1 + diaryTP +
                            Training_Condition +
               (1 | ID_numeric),
                      data =  child_abd_d2d3_longformat)

anova(model_c_time_asb, model_c_time_simple)
```

```
## refitting model(s) with ML (instead of REML)

## Data: child_abd_d2d3_longformat
## Models:
## model_c_time_simple: A_sum_child ~ ABD_sum_D1 + diaryTP + Training_Condition + (1 | ID_numeric)
## model_c_time_asb: A_sum_child ~ ABD_sum_D1 + diaryTP + Training_Condition + diaryTP * Training_Condi
##                       npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_c_time_simple    6 1208.5 1226.4 -598.25    1196.5
## model_c_time_asb       7 1210.5 1231.3 -598.24    1196.5 0.0348  1     0.8519
```

2.3) Simple model without interaction

```
model_c_time_simple <- lmer(A_sum_child ~ ABD_sum_D1 + diaryTP +
                            Training_Condition +
               (1 | ID_numeric),
                      data =  child_abd_d2d3_longformat)
```

2.3.1) Results

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_sum_child ~ ABD_sum_D1 + diaryTP + Training_Condition + (1 |
##     ID_numeric)
##    Data: child_abd_d2d3_longformat
##
## REML criterion at convergence: 1189.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9684 -0.3028 -0.1268  0.2793  4.2308
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ID_numeric (Intercept) 130.1    11.41
##  Residual               134.7    11.61
## Number of obs: 145, groups:  ID_numeric, 73
##
## Fixed effects:
##                          Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)               5.22183    2.96550 86.00068   1.761   0.0818 .
```

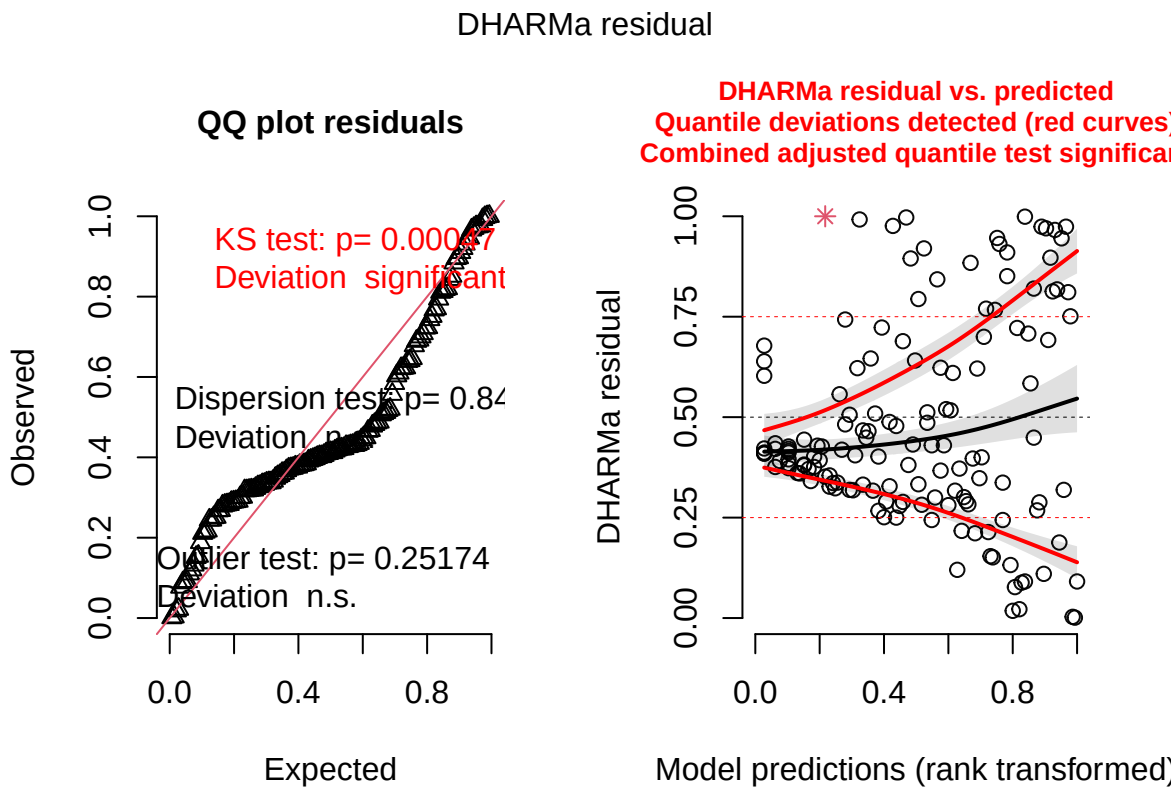
```
## ABD_sum_D1               0.63994    0.06988 71.06987   9.158 1.19e-13 ***
## diaryTP3                -0.75545    1.93142 71.47622  -0.391   0.6969
## Training_ConditionTRAIN -0.61318    3.32023 69.83766  -0.185   0.8540
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) ABD__D dryTP3
## ABD_sum_D1  -0.577
## diaryTP3    -0.326  0.012
## Trnn_CTRAIN -0.550  0.081 -0.004

## Type III Analysis of Variance Table with Satterthwaite's method
##                     Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## ABD_sum_D1         11299.6 11299.6     1 71.070 83.8618 1.191e-13 ***
## diaryTP               20.6    20.6     1 71.476  0.1530    0.6969
## Training_Condition     4.6     4.6     1 69.838  0.0341    0.8540
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Computing profile confidence intervals ...

##                              2.5 %     97.5 %
## .sig01                   8.1238248 14.2412660
## .sigma                   9.8734993 13.6970266
## (Intercept)             -0.5405514 10.9828957
## ABD_sum_D1               0.5041133  0.7759177
## diaryTP3                -4.5712368  3.0475171
## Training_ConditionTRAIN -7.0720736  5.8428720
```

2.3.2) Assumption Check



2.4) UPDATED FROM PREREGISTRATION: LMM Model predicting D3 only, adjusting for D1

Took out subject level random effect (as only one obs per ID) and added school. This gave singular fit. So ran LM model. Does it give singular fit?

```
model_c_time_asb_d3_only <- lmer(A_sum_child ~ ABD_sum_D1 +
                           Training_Condition +
               (1 | School.ID),
                      data =  child_abd_d3_only)

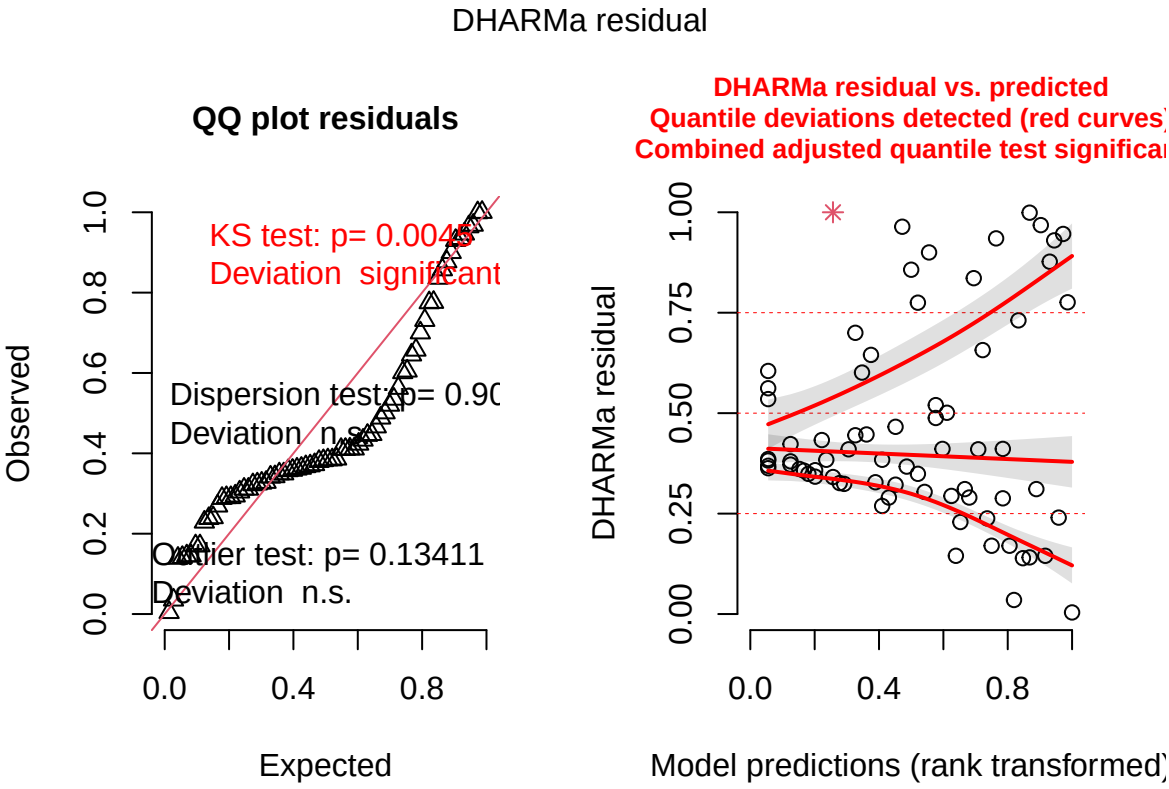


summary(model_c_time_asb_d3_only)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_sum_child ~ ABD_sum_D1 + Training_Condition + (1 | School.ID)
##    Data: child_abd_d3_only
##
## REML criterion at convergence: 608.7
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2744 -0.4590 -0.2083  0.1438  3.7898
##
## Random effects:
##  Groups    Name        Variance Std.Dev.
##  School.ID (Intercept)  30.75    5.546
##  Residual              285.19   16.888
## Number of obs: 72, groups:  School.ID, 15
##
## Fixed effects:
##                          Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)               5.61901    3.79466 40.85479   1.481    0.146
## ABD_sum_D1                0.60233    0.08886 68.60700   6.778 3.43e-09 ***
## Training_ConditionTRAIN -0.57093    4.05015 64.69856  -0.141    0.888
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) ABD__D
## ABD_sum_D1  -0.531
## Trnn_CTRAIN -0.525  0.072
```

2.4.1) Assumption Check



2.5) Bootstrapped LMM Model: Singular fit error given on many iterations

Bootstrapping:

```
# Fit your model first


# REB bootstrap advised when assumptions are violated
set.seed(123)  # For reproducibility
boot_results <- lmeresampler::bootstrap(model_c_time_asb_d3_only, .f = fixef, type = "reb", B = 5000)

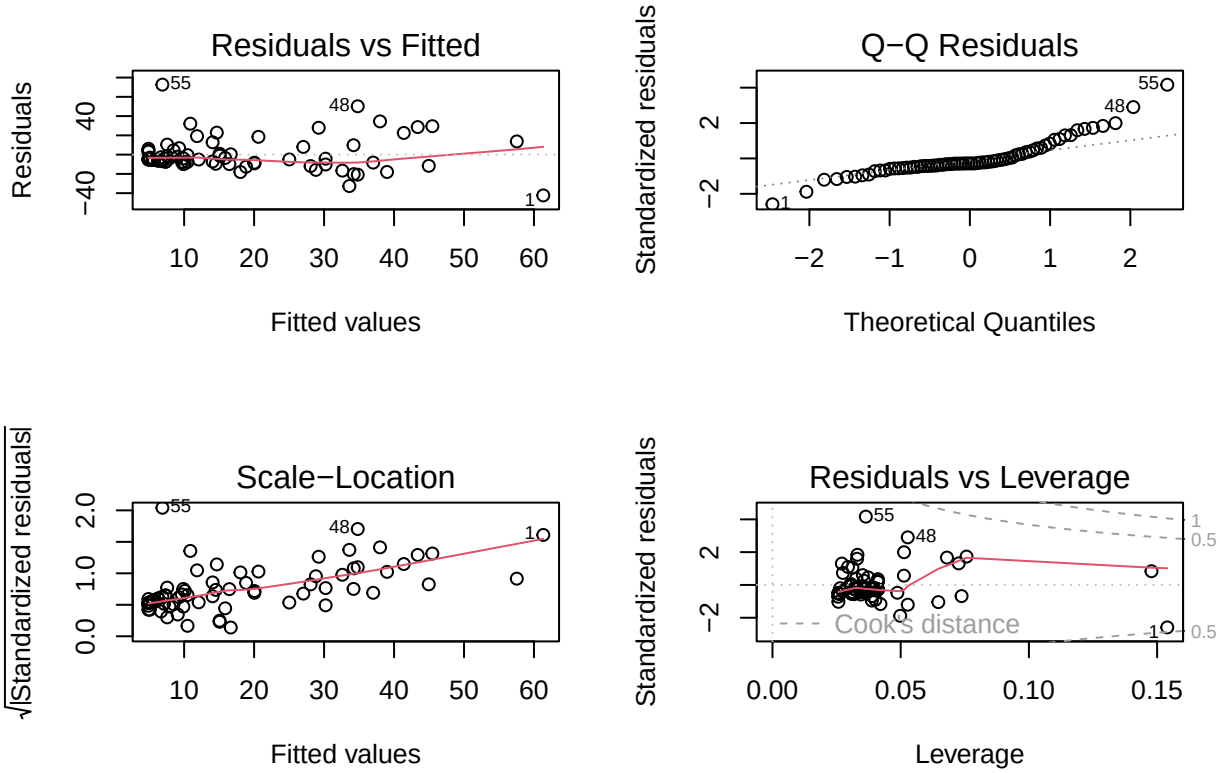
boot_results
```

```
## Bootstrap type: reb0
##
## Number of resamples: 5000
##
##                       term   observed   rep.mean         se         bias
## 1              (Intercept)  5.6190051  4.4663494 3.91623079 -1.152655681
## 2               ABD_sum_D1  0.6023295  0.6036354 0.08283341  0.001305963
## 3 Training_ConditionTRAIN -0.5709301 -0.5663975 3.71209054  0.004532537
##
## There were 441 messages, 0 warnings, and 0 errors.
##
## The most commonly occurring message was: boundary (singular) fit: see help('isSingular')
```

2.6) Linear Model

```
##
## Call:
## lm(formula = A_sum_child ~ ABD_sum_D1 + Training_Condition, data = child_abd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -42.328  -8.284  -4.902   4.906  72.657
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 5.0986     3.5444   1.438    0.155
## ABD_sum_D1                  0.5982     0.0902   6.632 6.13e-09 ***
## Training_ConditionTRAIN    -0.1961     4.2106  -0.047    0.963
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.77 on 69 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.3905, Adjusted R-squared:  0.3728
## F-statistic:  22.1 on 2 and 69 DF,  p-value: 3.82e-08
```

2.6.1) Assumption Check



2.7) Final Model: Bootstrapped Linear Model

```
boot_coef_c_asb_t3 <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(A_sum_child ~ ABD_sum_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb_t3 <- boot(data = child_abd_d3_only,
                      statistic = boot_coef_c_asb_t3,
                      R = 5000)
```

2.7.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_abd_d3_only, statistic = boot_coef_c_asb_t3,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1*  5.0985519  0.031741951   3.6553049
## t2*  0.5981868  0.002795416   0.1237681
## t3* -0.1961436 -0.130720014   4.1128521
```

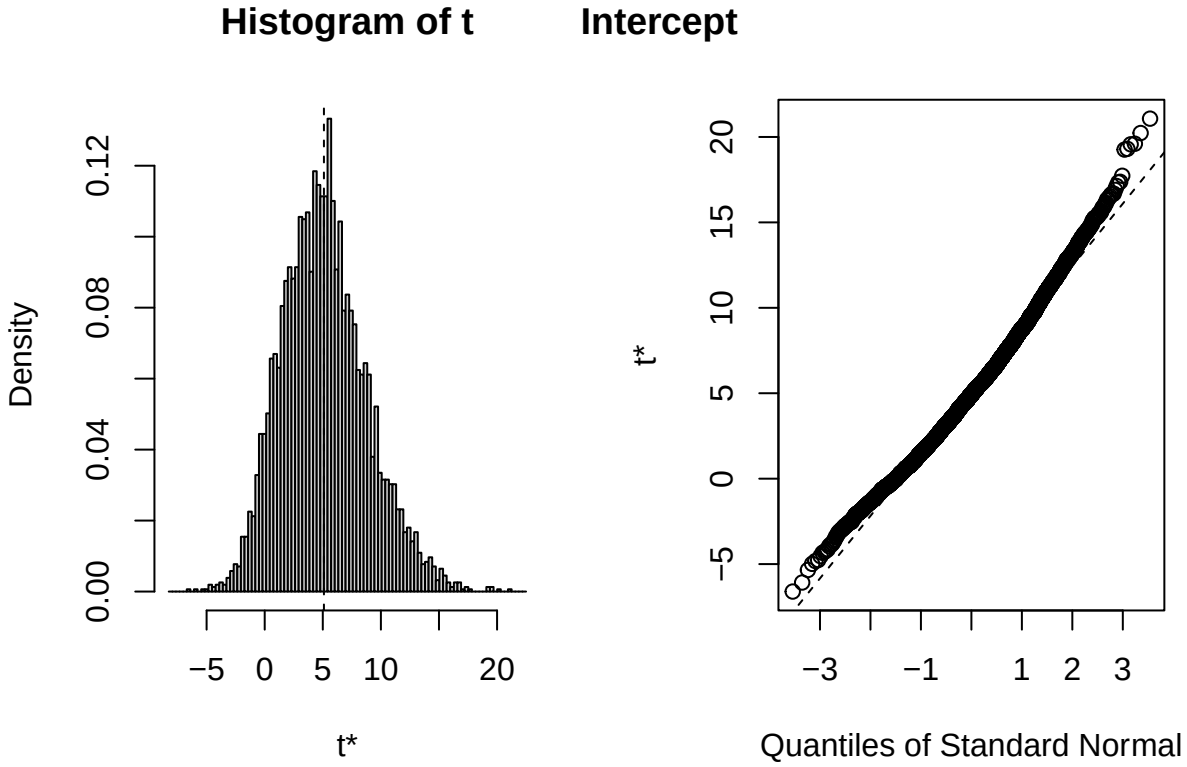
2.7.2) Confidence Intervals

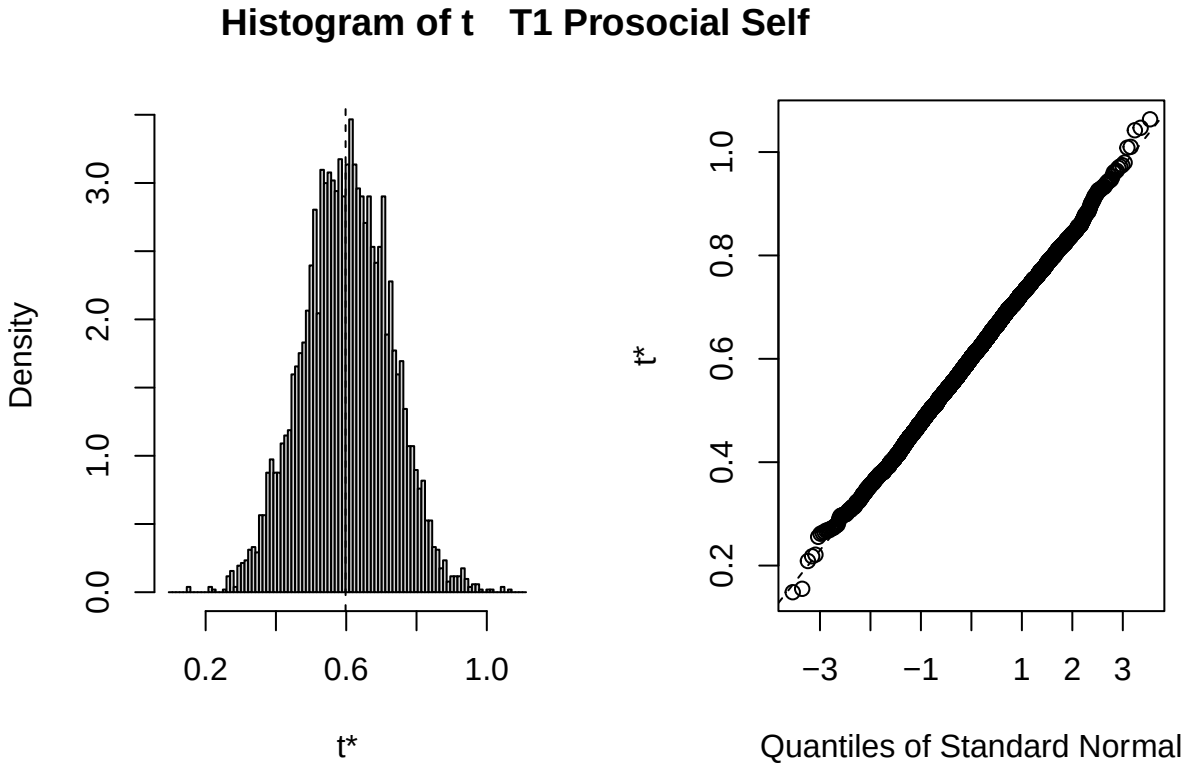
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_t3, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.354, 15.095 )
## Calculations and Intervals on Original Scale

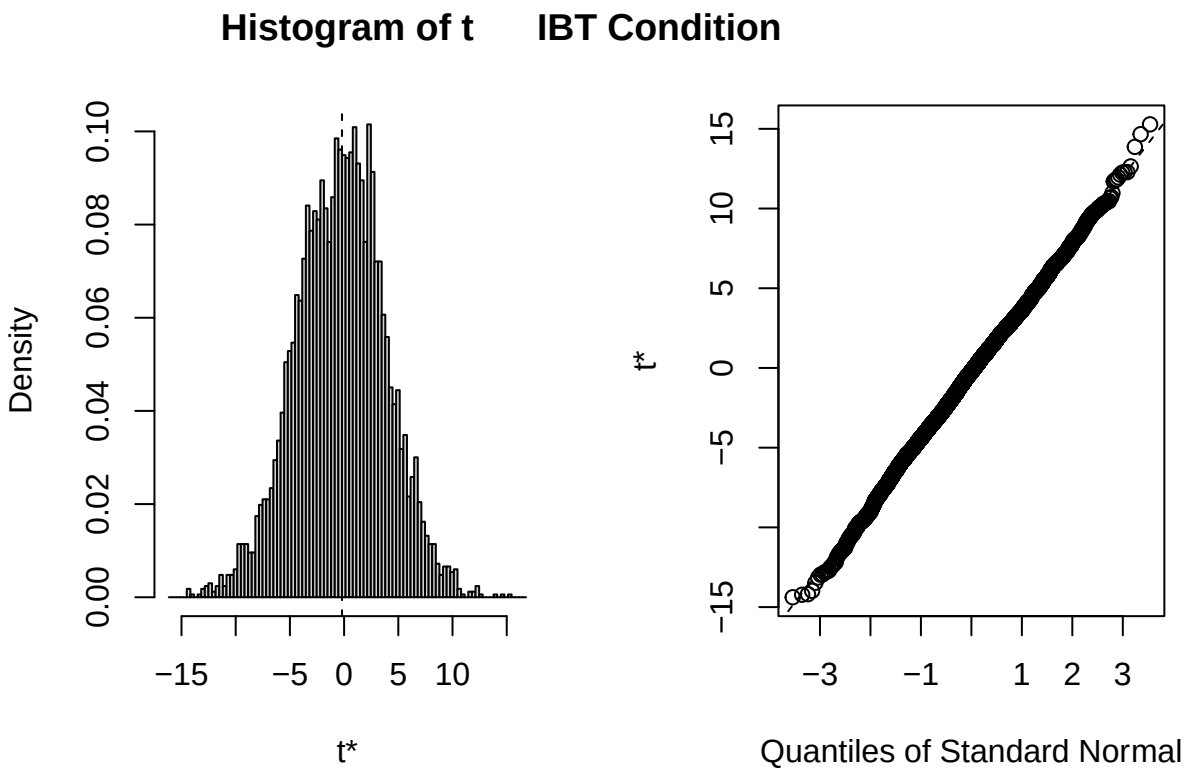
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_t3, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3382,  0.8179 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_t3, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-8.9369,  7.5015 )
## Calculations and Intervals on Original Scale
```

2.7.3) Bootstrap Diagnostics

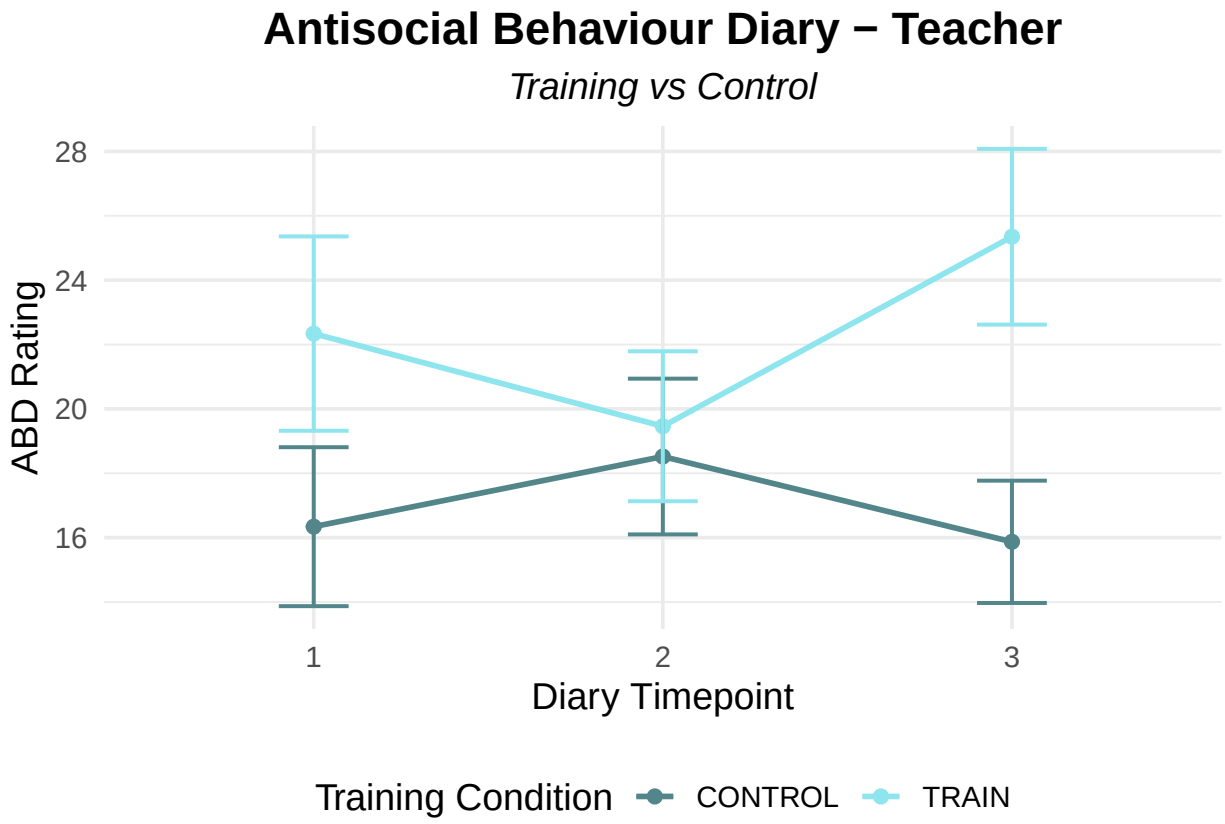





Histogram of t IBT Condition
Density
0.00
0.02
0.04
0.06
0.08
0.10
−15
−5
0
5
10
t*
t*
15
10
5
0
−5
−15
−3
−1
1
2
3
Quantiles of Standard Normal

3) Impact of training on teacher ratings of child antisocial at two months (T3)

3.1) Plot of teach ASB ratings over time



2.2) Prespecified LMM (without random slopes)

Trying prespecified model. This has the same issue as above (so removing the school random slope is correct):

```
# model that gives error:

# model_c_time_asb <- lmer(ABD_mean_rating ~ Mean_ABD_D1 + diaryTP +
#                            Training_Condition + diaryTP*Training_Condition +
#                          (1 + diaryTP|ID_numeric),
#                          data =  teach_abd_d2d3_longformat)


model_t_time_asb <- lmer(ABD_mean_rating ~ Mean_ABD_D1 + diaryTP +
                           Training_Condition + diaryTP*Training_Condition +
                         (1|ID_numeric),
                         data =  teach_abd_d2d3_longformat)


summary(model_t_time_asb)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: ABD_mean_rating ~ Mean_ABD_D1 + diaryTP + Training_Condition +
##     diaryTP * Training_Condition + (1 | ID_numeric)
##    Data: teach_abd_d2d3_longformat
##
## REML criterion at convergence: 1501.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.05158 -0.48608 -0.04694  0.38410  2.46336
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ID_numeric (Intercept) 96.53    9.825
##  Residual               99.26    9.963
## Number of obs: 190, groups:  ID_numeric, 95
##
## Fixed effects:
##                                  Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                       11.4309     2.2429 133.8078   5.096 1.16e-06
## Mean_ABD_D1                        0.4336     0.0668  92.0000   6.491 4.25e-09
## diaryTP3                          -2.6503     1.9730  93.0000  -1.343  0.18243
## Training_ConditionTRAIN           -1.6581     2.9069 147.0385  -0.570  0.56927
## diaryTP3:Training_ConditionTRAIN   8.5367     2.8990  93.0000   2.945  0.00408
##
## (Intercept)                      ***
## Mean_ABD_D1                      ***
## diaryTP3
## Training_ConditionTRAIN
## diaryTP3:Training_ConditionTRAIN **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##             (Intr) M_ABD_ dryTP3 T_CTRA
## Mean_ABD_D1 -0.487
## diaryTP3    -0.440  0.000
## Trnn_CTRAIN -0.522 -0.138  0.339
## dTP3:T_CTRA  0.299  0.000 -0.681 -0.499
```

```
anova(model_t_time_asb)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                            Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## Mean_ABD_D1                4182.2  4182.2     1    92 42.1339 4.248e-09 ***
## diaryTP                     123.7   123.7     1    93  1.2460  0.267192
## Training_Condition          106.5   106.5     1    92  1.0732  0.302945
## diaryTP:Training_Condition  860.7   860.7     1    93  8.6711  0.004084 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
confint(model_t_time_asb)
```

```
## Computing profile confidence intervals ...
```

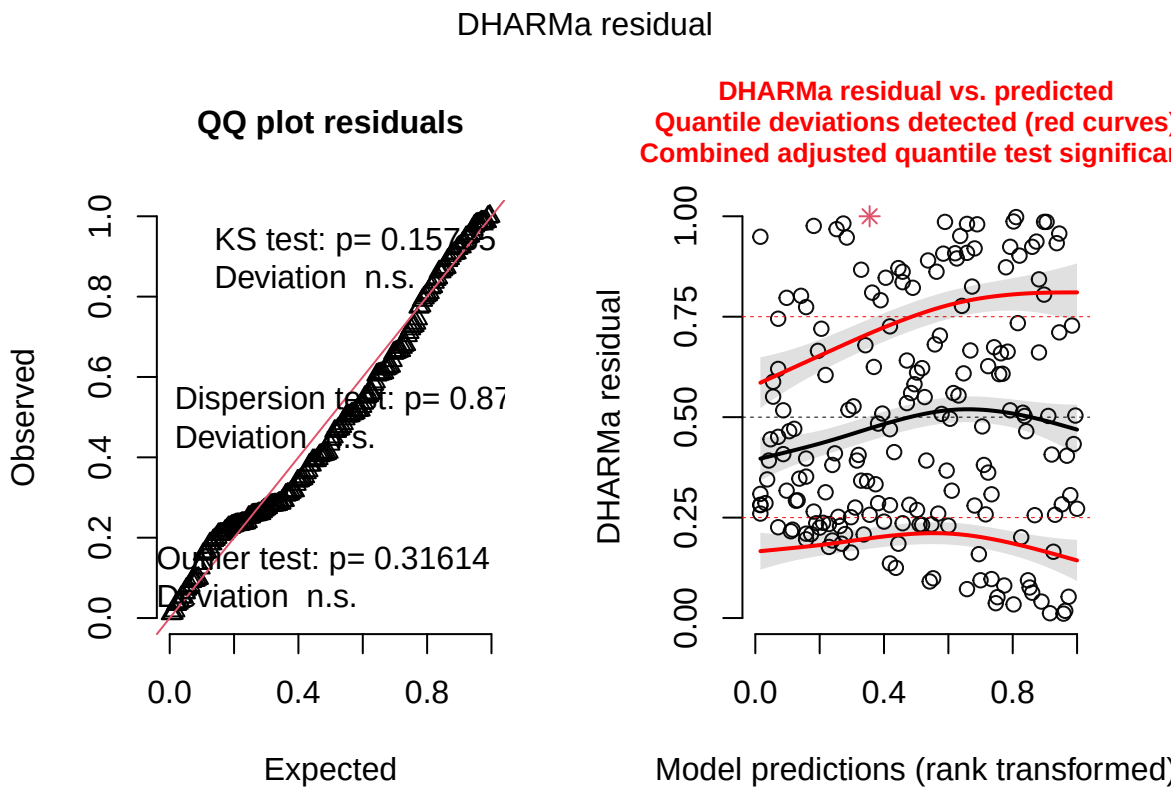
```
##                                      2.5 %     97.5 %
## .sig01                           7.4515928 11.9861193
## .sigma                           8.6061343 11.4443386
## (Intercept)                      7.0699250 15.7918135
## Mean_ABD_D1                      0.3034672  0.5637944
## diaryTP3                        -6.5153311  1.2146775
## Training_ConditionTRAIN         -7.3081257  3.9919319
## diaryTP3:Training_ConditionTRAIN  2.8575128 14.2158680
```

```
confint(model_t_time_asb, level = 0.995)
```

```
## Computing profile confidence intervals ...
```

```
##                                     0.25 %    99.75 %
## .sig01                           6.4211996 13.1243864
## .sigma                           8.1459709 12.2642679
## (Intercept)                      5.1385180 17.7232205
## Mean_ABD_D1                      0.2452007  0.6220609
## diaryTP3                        -8.2454638  2.9448102
## Training_ConditionTRAIN         -9.8048865  6.4886926
## diaryTP3:Training_ConditionTRAIN  0.3152826 16.7580981
```

2.2.1) Assumption Check



2.2.2) Complexity Check - a model WITH an interaction is better fit.

```
## refitting model(s) with ML (instead of REML)

## Data: teach_abd_d2d3_longformat
## Models:
## model_t_asb_no_interaction: ABD_mean_rating ~ Mean_ABD_D1 + diaryTP + Training_Condition + (1 | ID_nu
## model_t_time_asb: ABD_mean_rating ~ Mean_ABD_D1 + diaryTP + Training_Condition + diaryTP * Training_C
##                            npar    AIC    BIC  logLik -2*log(L)  Chisq Df
## model_t_asb_no_interaction    6 1530.8 1550.3 -759.39    1518.8
## model_t_time_asb              7 1524.3 1547.0 -755.16    1510.3 8.4686  1
##                            Pr(>Chisq)
## model_t_asb_no_interaction
## model_t_time_asb             0.003613 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2.2.3) Boostrapped preregistered LMM Model

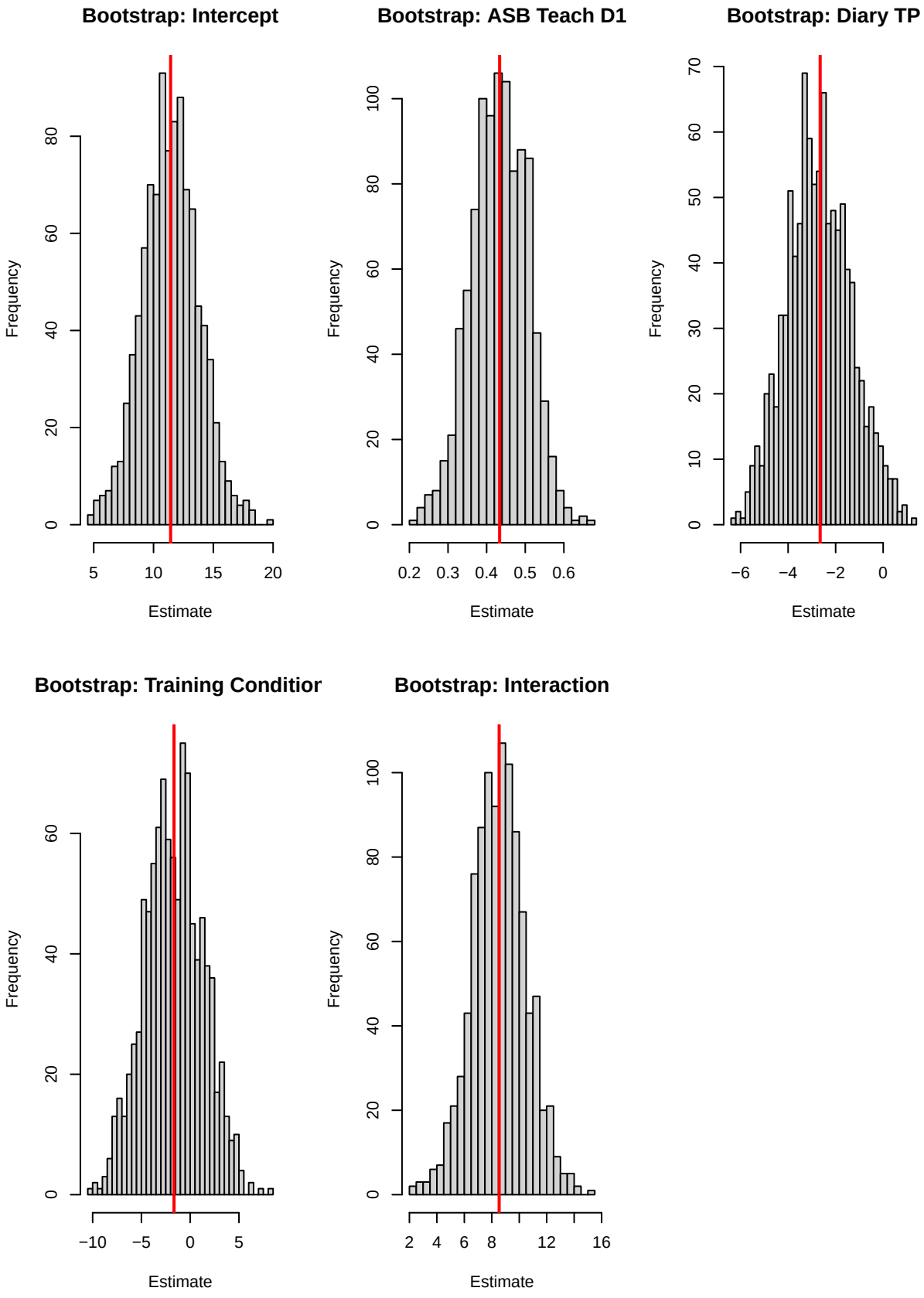
```
# Fit your model first
# REB bootstrap advised when assumptions are violated
set.seed(123)  # For reproducibility
boot_results <- lmeresampler::bootstrap(model_t_time_asb, .f = fixef, type = "reb", B = 1000)
```

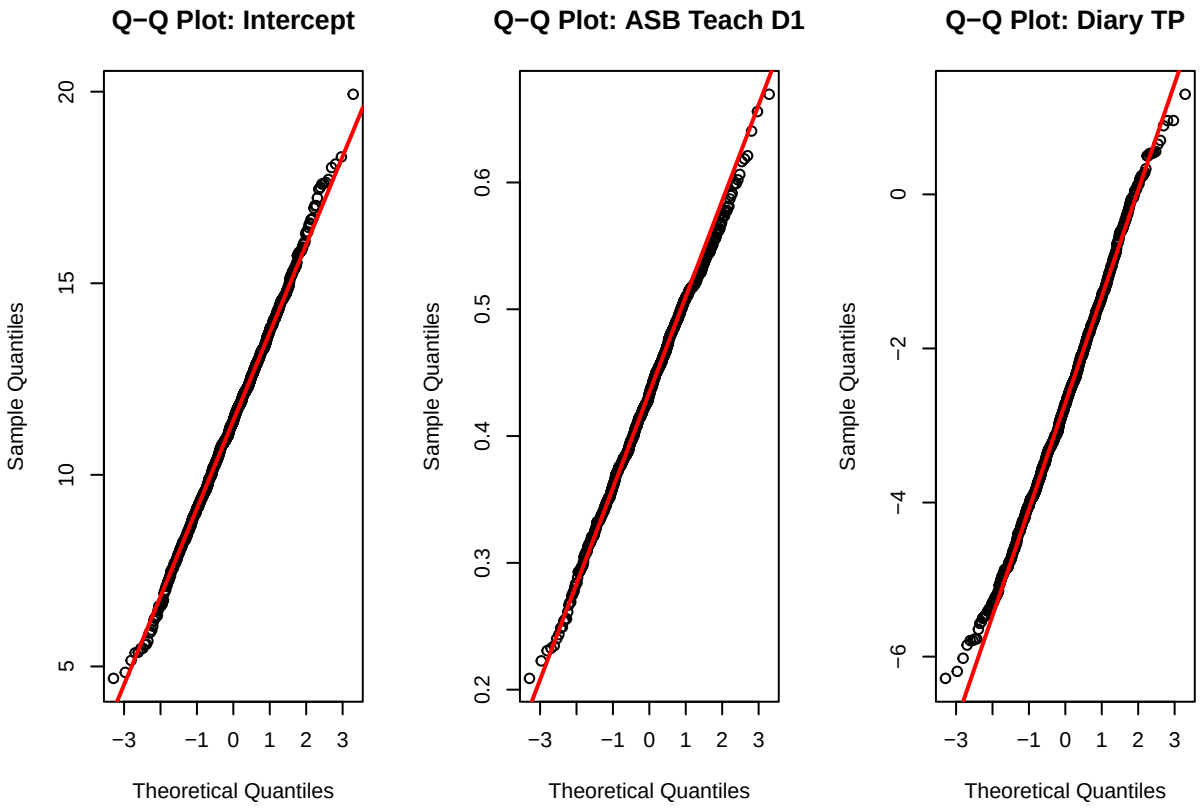
2.2.4) Bootstrap LMM results and confidence intervals

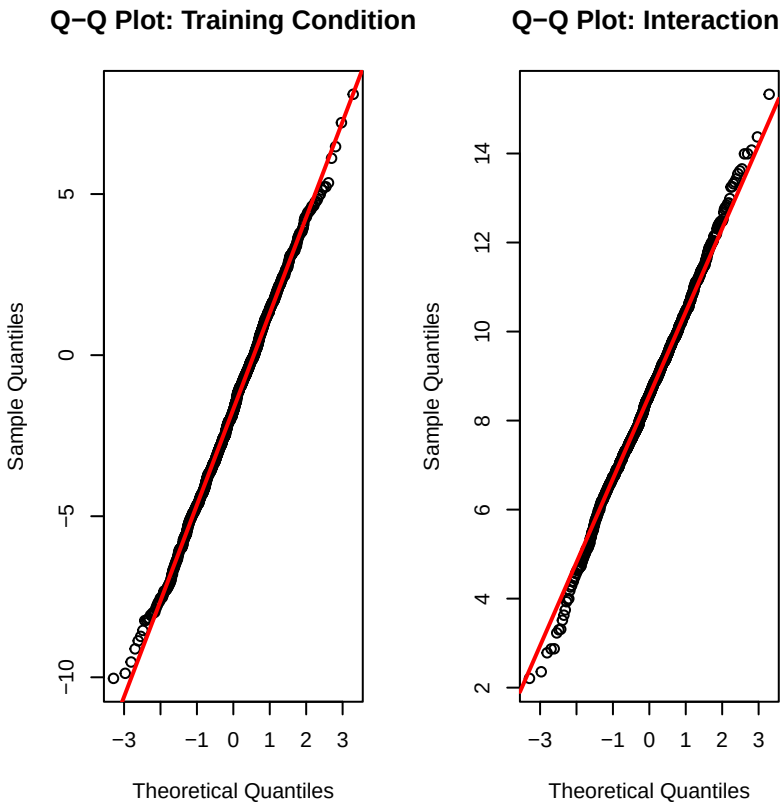
```
## # A tibble: 5 x 6
##   term                              estimate  lower  upper type  level
##   <chr>                                <dbl>  <dbl>  <dbl> <chr> <dbl>
## 1 (Intercept)                          11.4    4.92  18.3   perc  0.998
## 2 Mean_ABD_D1                           0.434  0.225  0.652 perc  0.998
## 3 diaryTP3                             -2.65  -6.15   0.958 perc  0.998
## 4 Training_ConditionTRAIN              -1.66  -9.79   7.03  perc  0.998
## 5 diaryTP3:Training_ConditionTRAIN      8.54   2.46  14.3   perc  0.998

## Bootstrap type: reb0
##
## Number of resamples: 1000
##
##                                term   observed  rep.mean         se
## 1                       (Intercept) 11.4308693 11.427948 2.35513261
## 2                       Mean_ABD_D1  0.4336308  0.433150 0.07229213
## 3                          diaryTP3 -2.6503268 -2.690852 1.35256097
## 4          Training_ConditionTRAIN -1.6580969 -1.668975 2.97635876
## 5 diaryTP3:Training_ConditionTRAIN  8.5366904  8.562471 1.95931800
##             bias
## 1 -0.0029208704
## 2 -0.0004808008
## 3 -0.0405249846
## 4 -0.0108783369
## 5  0.0257800908
##
## There were 0 messages, 0 warnings, and 0 errors.
```

2.2.5) Bootstrap LMM diagnostics




Q−Q Plot: Intercept
Sample Quantiles
Theoretical Quantiles
5
10
15
20
−3
−1
0
1
2
3
Q−Q Plot: ASB Teach D1
Sample Quantiles
Theoretical Quantiles
0.2
0.3
0.4
0.5
0.6
Q−Q Plot: Diary TP
Sample Quantiles
Theoretical Quantiles
−6
−4
−2
0



2.4) UPDATED FROM PREREGISTRATION: LMM Predicting D3 only, adusting for D1 - this gives singular fit

Added School as random effect not ID.

```
model_t_time_asb_nod2_full<- lmer(ABD_mean_rating ~ Mean_ABD_D1 +
                          Training_Condition +
                       (1|School.ID/teachUUID_diaries),
                       data =  teach_abd_d3_only)
```

```
## boundary (singular) fit: see help('isSingular')
```

2.5) LM Predicting D3 only, adusting for D1

```
## lmer gives singular fit so trying simple lm model
model_t_time_asb_nod2_lm<- lm(ABD_mean_rating ~ Mean_ABD_D1 +
                          Training_Condition,
                       data =  teach_abd_d3_only)

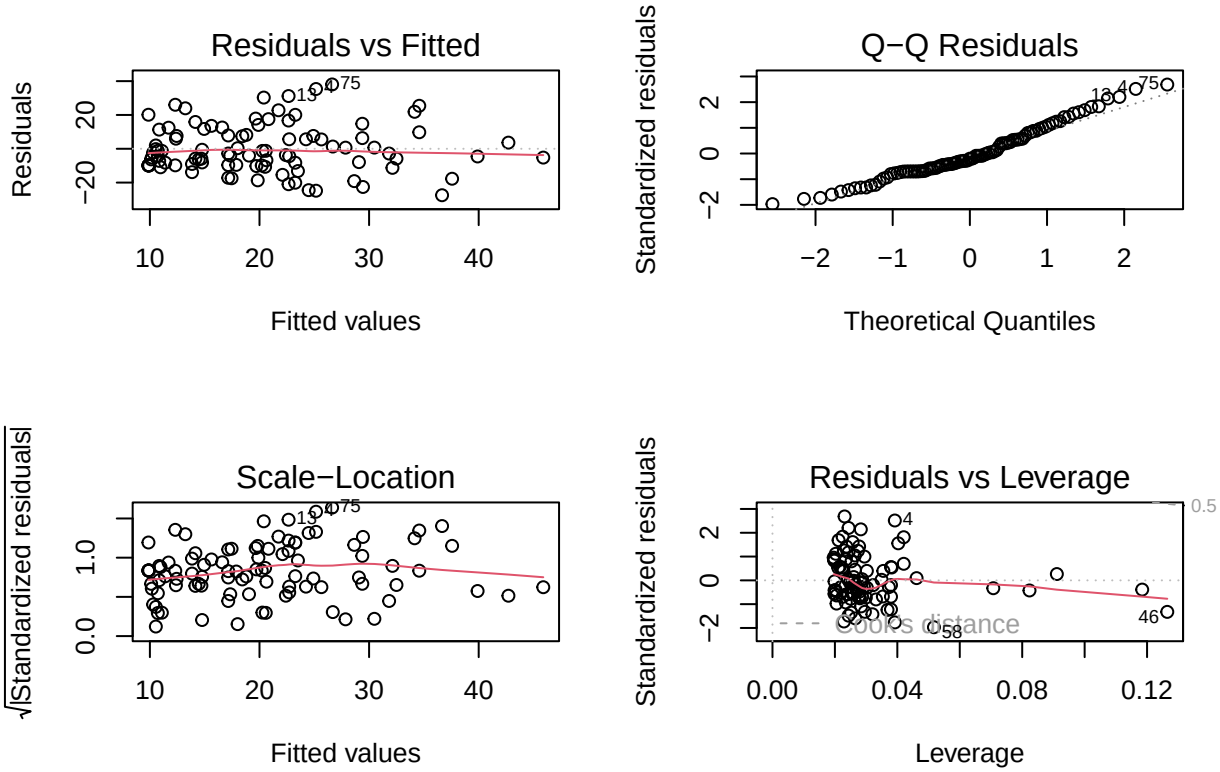
summary(model_t_time_asb_nod2_lm)
```

```
##
```

```
## Call:
## lm(formula = ABD_mean_rating ~ Mean_ABD_D1 + Training_Condition,
##     data = teach_abd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.492  -9.843  -2.827   8.026  38.038
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               9.87032    2.38826   4.133 7.89e-05 ***
## Mean_ABD_D1               0.36694    0.07921   4.633 1.18e-05 ***
## Training_ConditionTRAIN  7.27908    2.98744   2.437   0.0168 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.33 on 92 degrees of freedom
## Multiple R-squared:  0.2569, Adjusted R-squared:  0.2407
## F-statistic:  15.9 on 2 and 92 DF,  p-value: 1.173e-06
```

2.5.1) Assumption Check

Checking residuals of simple model: (commen)



2.6) Final Model: Bootstrapped linear regression

Teacher full data bootstrapping antisocial

```
# non-parametric case bootstrap: https://journal.r-project.org/articles/RN-2002-017/RN-2002-017.pdf

boot_coef_t_asb_D3 <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ABD_mean_rating ~ Mean_ABD_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(1273)  # For reproducibility
boot_results_t_asb_D3 <- boot(data = teach_abd_d3_only,
                      statistic = boot_coef_t_asb_D3,
                      R = 5000)
```

2.6.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_abd_d3_only, statistic = boot_coef_t_asb_D3,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original       bias    std. error
## t1* 9.8703160 -0.024822716   1.8743537
## t2* 0.3669367  0.003556156   0.0786095
## t3* 7.2790804 -0.013566153   3.0728055
```

2.6.2) Confidence Intervals - 95%

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_D3, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 6.350, 13.817 )
## Calculations and Intervals on Original Scale
```

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_D3, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.2244,  0.5299 )
## Calculations and Intervals on Original Scale
```

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_D3, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   ( 1.305, 13.334 )
## Calculations and Intervals on Original Scale
```

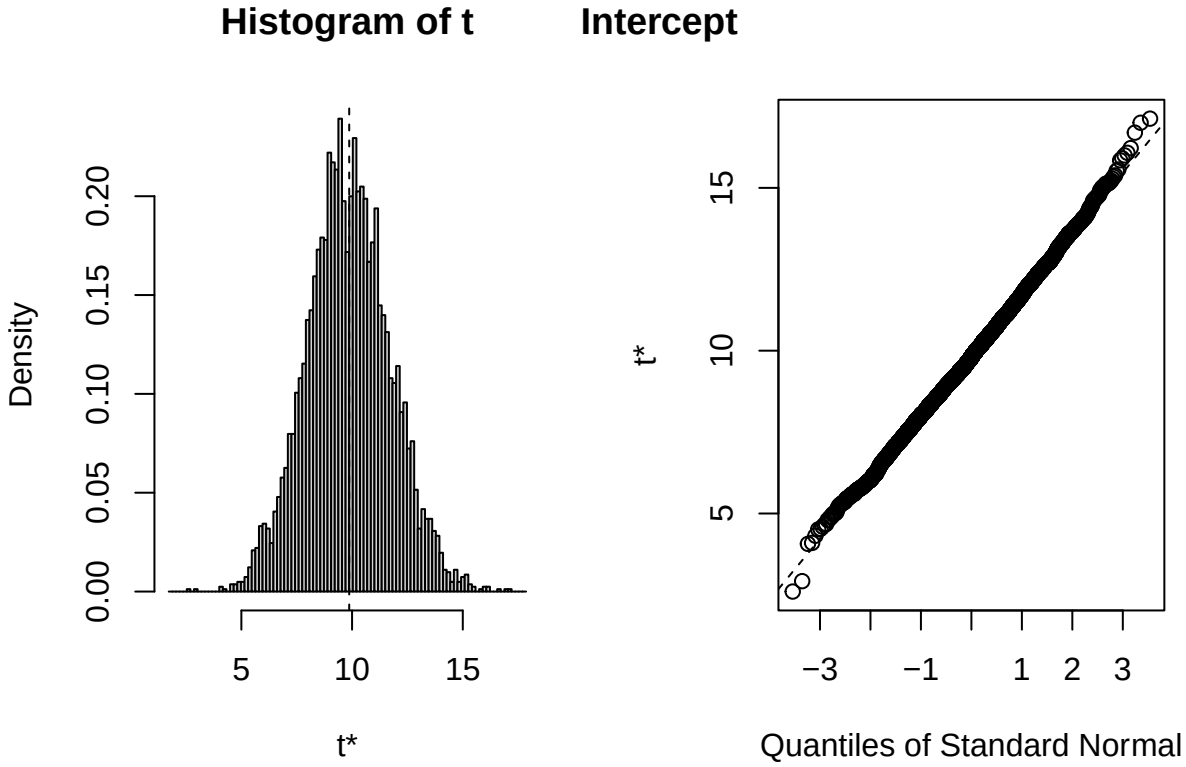
2.6.3) Confidence Intervals - 99%

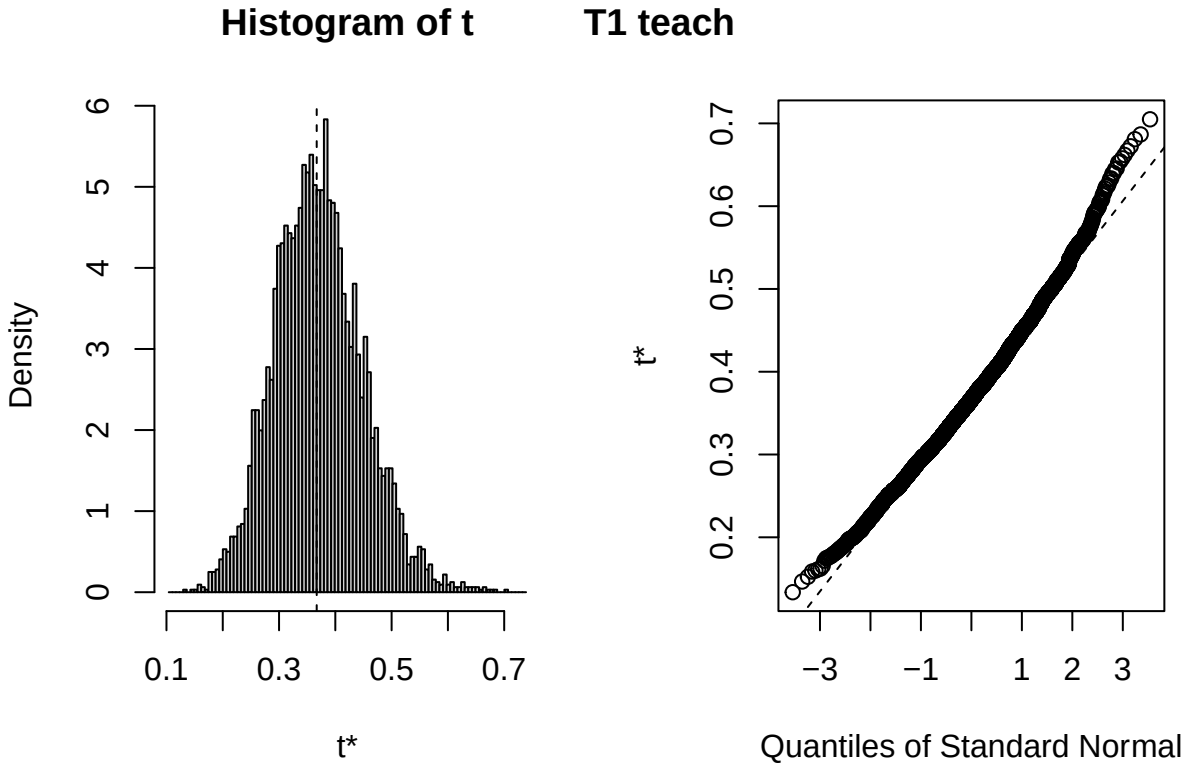
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_D3, conf = 0.99, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 99%   ( 5.499, 15.165 )
## Calculations and Intervals on Original Scale

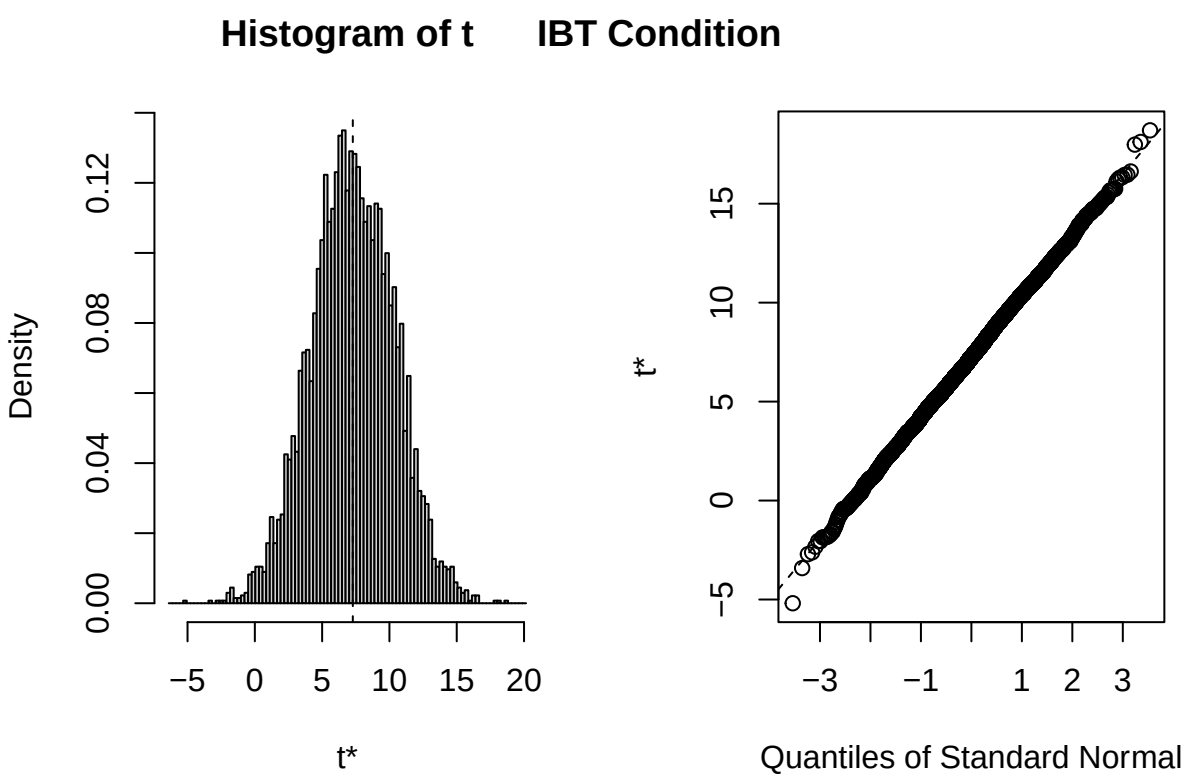
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_D3, conf = 0.99, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 99%   ( 0.1841,  0.5989 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_D3, conf = 0.99, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 99%   (-0.410, 15.281 )
## Calculations and Intervals on Original Scale
```

2.6.4) Model Diagnostics

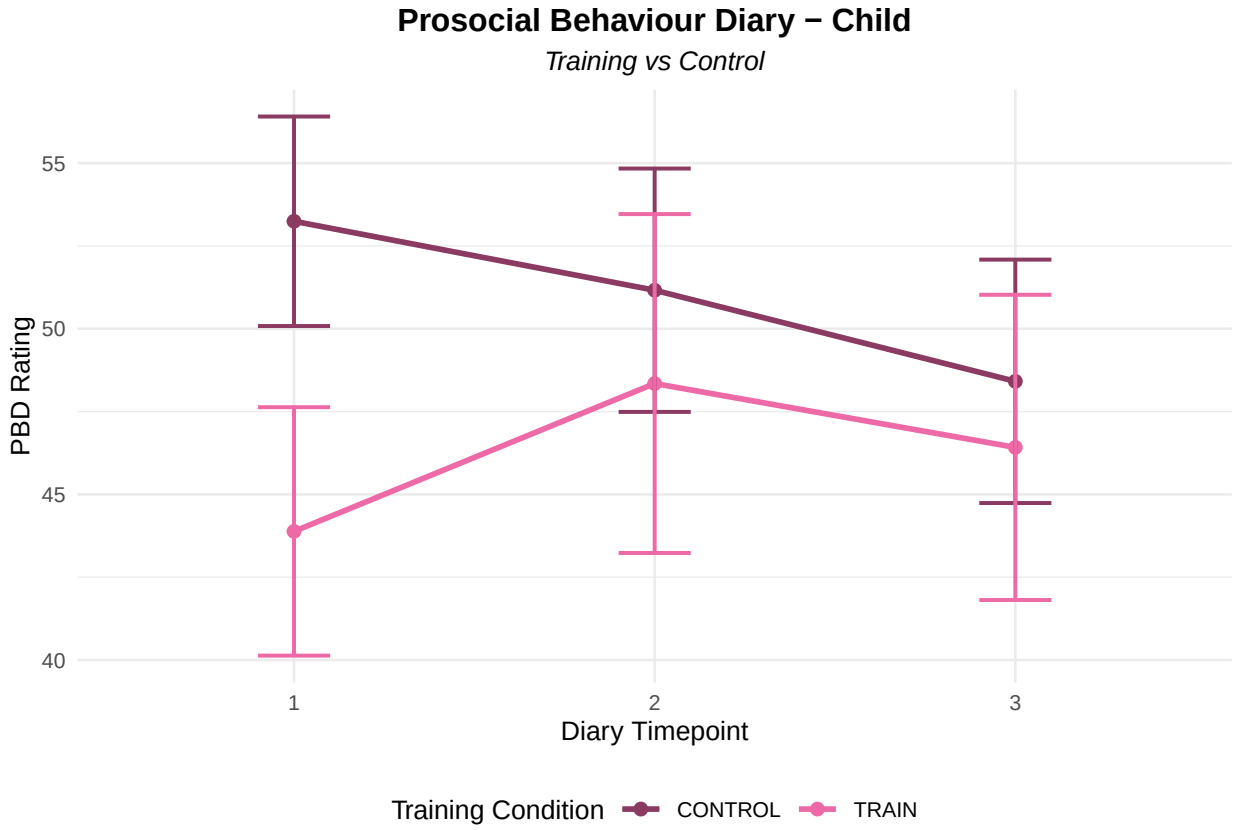




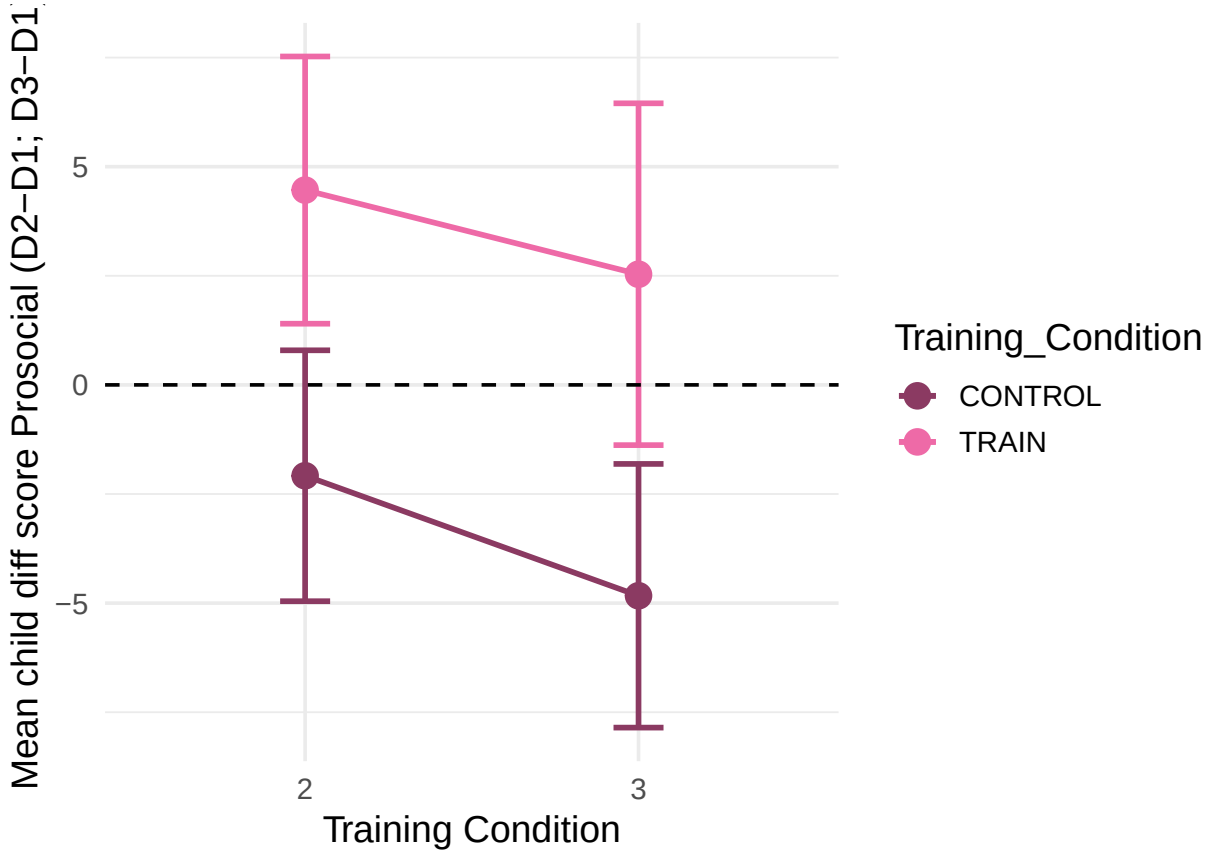

Histogram of t IBT Condition
Density
0.00
0.04
0.08
0.12
−5
0
5
10
15
20
t*
t*
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal

3) Impact of training over time on child ratings of prosocial behaviour at two months (T3)

3.1) Plotting child prosocial ratings over time



Difference plot:



2.2) Preregistered LMM (with now random slopes)

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: P_sum_child ~ PBD_sum_D1 + diaryTP + Training_Condition + diaryTP *
##     Training_Condition + (1 | ID_numeric)
##    Data: child_pbd_d2d3_longformat
##
## REML criterion at convergence: 1242.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.1375 -0.5284 -0.0251  0.3930  2.8947
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ID_numeric (Intercept) 229.4    15.14
##  Residual               141.9    11.91
## Number of obs: 148, groups:  ID_numeric, 74
##
## Fixed effects:
##                                 Estimate Std. Error        df t value
## (Intercept)                      7.38475    6.03412  78.25646   1.224
```

```
## PBD_sum_D1                          0.82221    0.09782  70.99999   8.405
## diaryTP3                           -2.74975    2.66379  72.00000  -1.032
## Training_ConditionTRAIN             4.88074    4.58695 101.44719   1.064
## diaryTP3:Training_ConditionTRAIN    0.82210    3.92986  72.00000   0.209
##                                   Pr(>|t|)
## (Intercept)                          0.225
## PBD_sum_D1                        2.96e-12 ***
## diaryTP3                             0.305
## Training_ConditionTRAIN              0.290
## diaryTP3:Training_ConditionTRAIN     0.835
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) PBD__D dryTP3 T_CTRA
## PBD_sum_D1  -0.863
## diaryTP3    -0.221  0.000
## Trnn_CTRAIN -0.508  0.200  0.290
## dTP3:T_CTRA  0.150  0.000 -0.678 -0.428
```

2.2.1) Complexity Check: Comparing model with interaction to one without. Interaction does not improve model.

```
model_c_time_psb_simple <- lmer(P_sum_child ~ PBD_sum_D1 + diaryTP +
                          Training_Condition +
                      (1|ID_numeric),
                      data =  child_pbd_d2d3_longformat)

anova(model_c_time_psb, model_c_time_psb_simple)
```

```
## refitting model(s) with ML (instead of REML)

## Data: child_pbd_d2d3_longformat
## Models:
## model_c_time_psb_simple: P_sum_child ~ PBD_sum_D1 + diaryTP + Training_Condition + (1 | ID_numeric)
## model_c_time_psb: P_sum_child ~ PBD_sum_D1 + diaryTP + Training_Condition + diaryTP * Training_Condi
##                          npar    AIC    BIC  logLik -2*log(L) Chisq Df
## model_c_time_psb_simple    6 1267.1 1285.1 -627.55    1255.1
## model_c_time_psb           7 1269.0 1290.0 -627.53    1255.0 0.045  1
##                          Pr(>Chisq)
## model_c_time_psb_simple
## model_c_time_psb            0.8321
```

2.3) LMM Without interaction

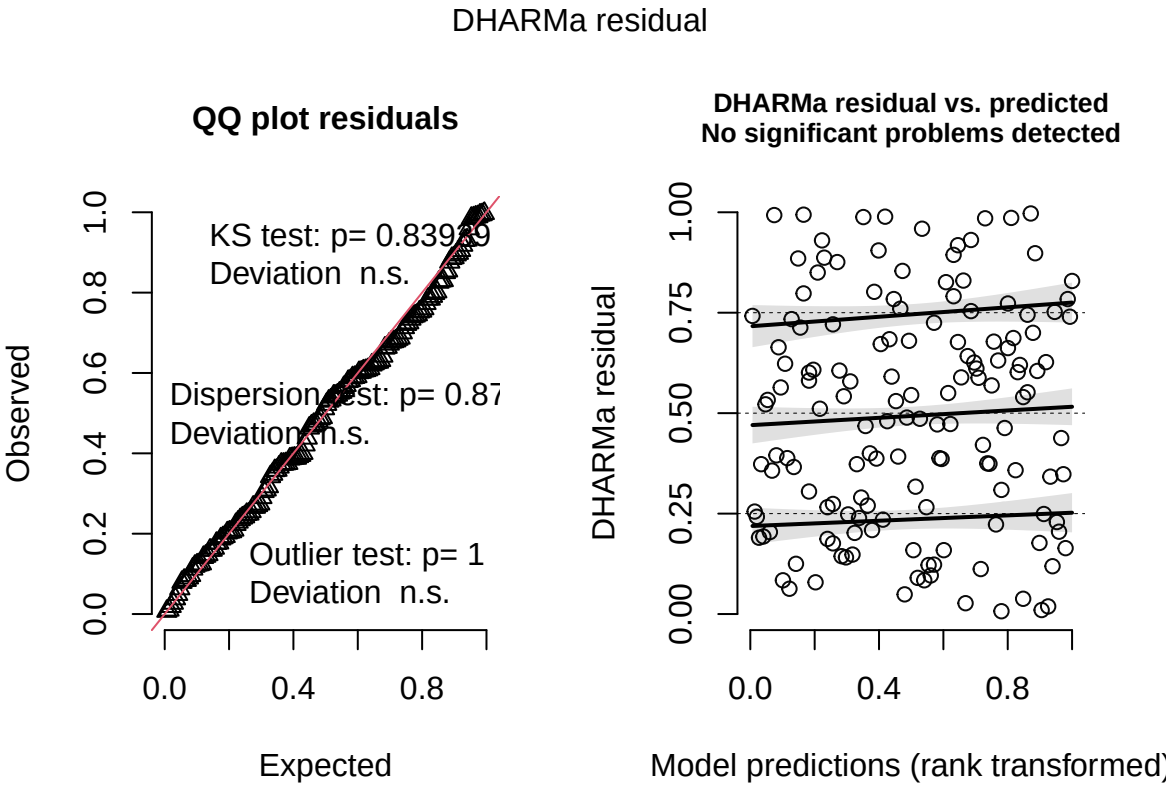
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: P_sum_child ~ PBD_sum_D1 + diaryTP + Training_Condition + (1 |
##     ID_numeric)
##    Data: child_pbd_d2d3_longformat
##
## REML criterion at convergence: 1246.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.17539 -0.53567 -0.02769  0.41110  2.92763
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ID_numeric (Intercept) 230.3    15.18
##  Residual               140.1    11.83
## Number of obs: 148, groups:  ID_numeric, 74
##
## Fixed effects:
##                          Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)               7.19589    5.96515 74.87830   1.206    0.231
## PBD_sum_D1                0.82221    0.09782 71.00000   8.405 2.96e-12 ***
## diaryTP3                 -2.37203    1.94559 73.00000  -1.219    0.227
## Training_ConditionTRAIN  5.29180    4.14477 71.00000   1.277    0.206
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) PBD__D dryTP3
## PBD_sum_D1  -0.873
## diaryTP3    -0.163  0.000
## Trnn_CTRAIN -0.497  0.221  0.000

## Type III Analysis of Variance Table with Satterthwaite's method
##                    Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## PBD_sum_D1         9894.9  9894.9     1    71 70.6494 2.962e-12 ***
## diaryTP             208.2   208.2     1    73  1.4864    0.2267
## Training_Condition  228.3   228.3     1    71  1.6301    0.2059
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Computing profile confidence intervals ...

##                              2.5 %    97.5 %
## .sig01                   11.7741255 18.371833
## .sigma                   10.0877411 13.935869
## (Intercept)              -4.4029038 18.794688
## PBD_sum_D1                0.6319503  1.012474
## diaryTP3                 -6.2091465  1.465092
## Training_ConditionTRAIN -2.7698268 13.353417
```

2.3.1) Assumption Checks



2.4) UPDATED FROM PREREGISTAION: model predicting D3 ratings, adjusting for D1

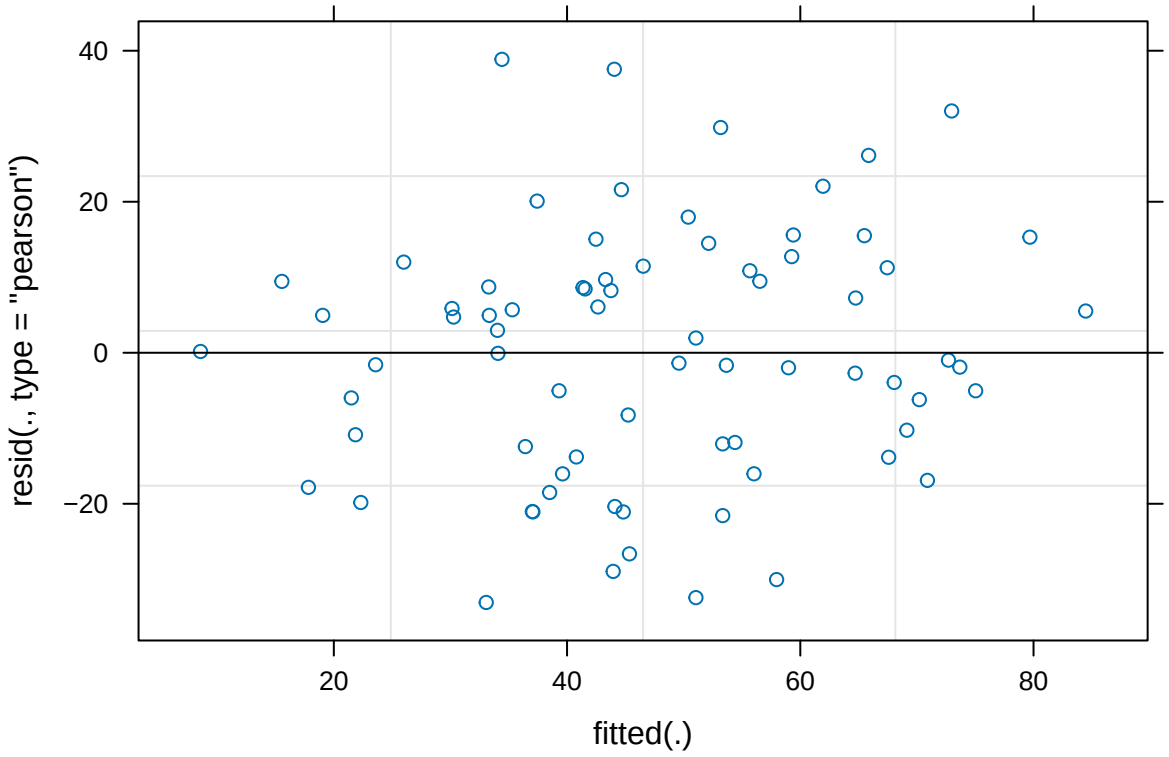
```
model_c_time_psb_d3only <- lmer(P_sum_child ~ PBD_sum_D1 +
                          Training_Condition +
                      (1|School.ID),
                      data =  child_pbd_d3_only)




summary(model_c_time_psb_d3only)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: P_sum_child ~ PBD_sum_D1 + Training_Condition + (1 | School.ID)
##    Data: child_pbd_d3_only
##
## REML criterion at convergence: 639.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.87042 -0.69775  0.00263  0.59787  2.19810
##
## Random effects:
##  Groups    Name        Variance Std.Dev.
##  School.ID (Intercept) 112.8    10.62
##  Residual              312.6    17.68
## Number of obs: 74, groups:  School.ID, 15
##
## Fixed effects:
##                          Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)               11.2076     7.0246 60.0935   1.595    0.116
## PBD_sum_D1                 0.7485     0.1079 68.0099   6.938 1.83e-09 ***
## Training_ConditionTRAIN    3.7863     4.3031 60.7463   0.880    0.382
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) PBD__D
## PBD_sum_D1  -0.805
## Trnn_CTRAIN -0.437  0.202
```

2.4.1) Assumption Check



2.5) Bootstrapped LMM - this returns singular fit

```
## Bootstrap type: reb0
##
## Number of resamples: 5000
##
##                       term   observed   rep.mean         se         bias
## 1              (Intercept) 11.2076391 12.5126619 7.23632418  1.305022851
## 2               PBD_sum_D1  0.7484646  0.7460895 0.09858051 -0.002375148
## 3 Training_ConditionTRAIN  3.7862849  3.7428335 3.77285204 -0.043451387
##
## There were 6 messages, 0 warnings, and 0 errors.
##
## The most commonly occurring message was: boundary (singular) fit: see help('isSingular')
```

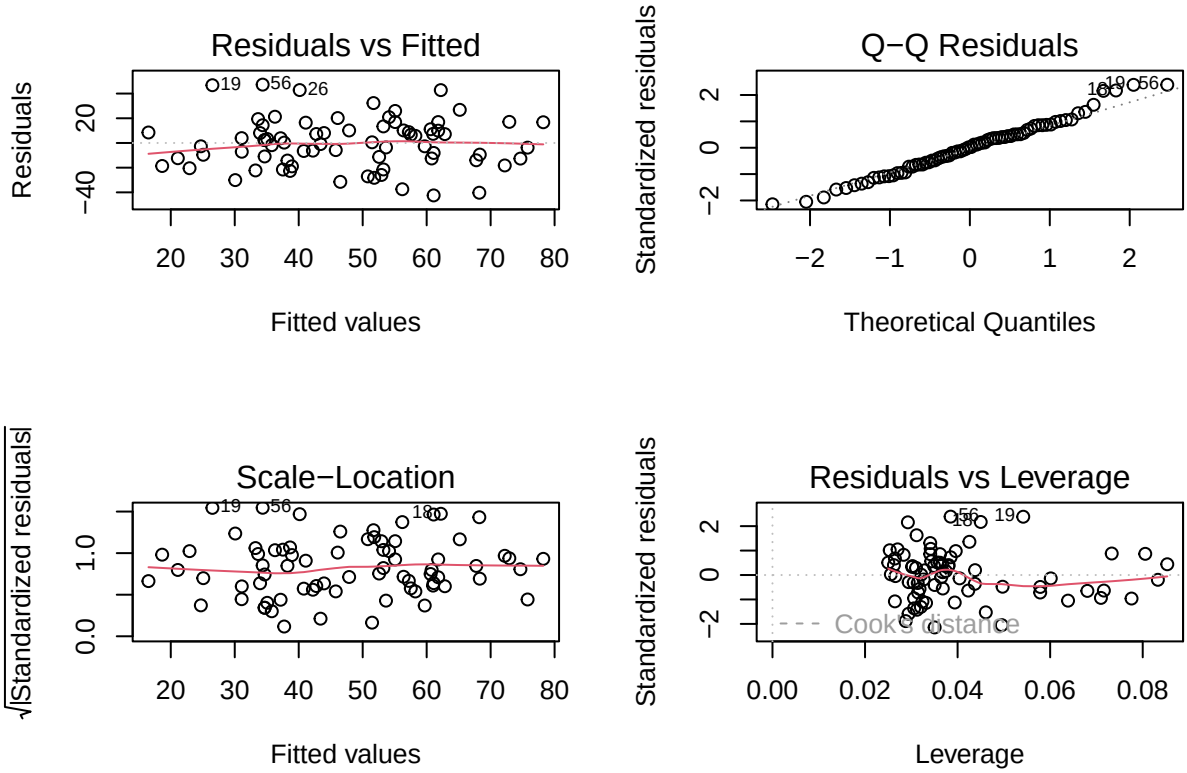
2.6) Simple Linear Regression Model

```
model_c_time_psb_lm <- lm(P_sum_child ~ PBD_sum_D1 +
                            Training_Condition,
                          data =  child_pbd_d3_only)

summary(model_c_time_psb_lm)
```

```
##
## Call:
## lm(formula = P_sum_child ~ PBD_sum_D1 + Training_Condition, data = child_pbd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -42.472 -12.636   0.403  10.867  47.237
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                10.3930     6.8489   1.517    0.134
## PBD_sum_D1                  0.7141     0.1138   6.273 2.46e-08 ***
## Training_ConditionTRAIN     4.6904     4.8234   0.972    0.334
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.17 on 71 degrees of freedom
## Multiple R-squared:  0.3576, Adjusted R-squared:  0.3395
## F-statistic: 19.76 on 2 and 71 DF,  p-value: 1.501e-07
```

2.6.1) Assumption Check



2.7) Final Model: Bootstrapped Linear Regression

```r
# non-parametric case bootstrap: https://journal.r-project.org/articles/RN-2002-017/RN-2002-017.pdf

boot_coef_c_psb_t3 <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(P_sum_child ~ PBD_sum_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(1223)  # For reproducibility
boot_results_c_psb_t3 <- boot(data = child_pbd_d3_only,
                      statistic = boot_coef_c_psb_t3,
                      R = 5000)
```

2.7.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_pbd_d3_only, statistic = boot_coef_c_psb_t3,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 10.3929948  0.171793948   5.8927382
## t2*  0.7140725 -0.003203682   0.1078812
## t3*  4.6903947 -0.083156445   4.6688996
```

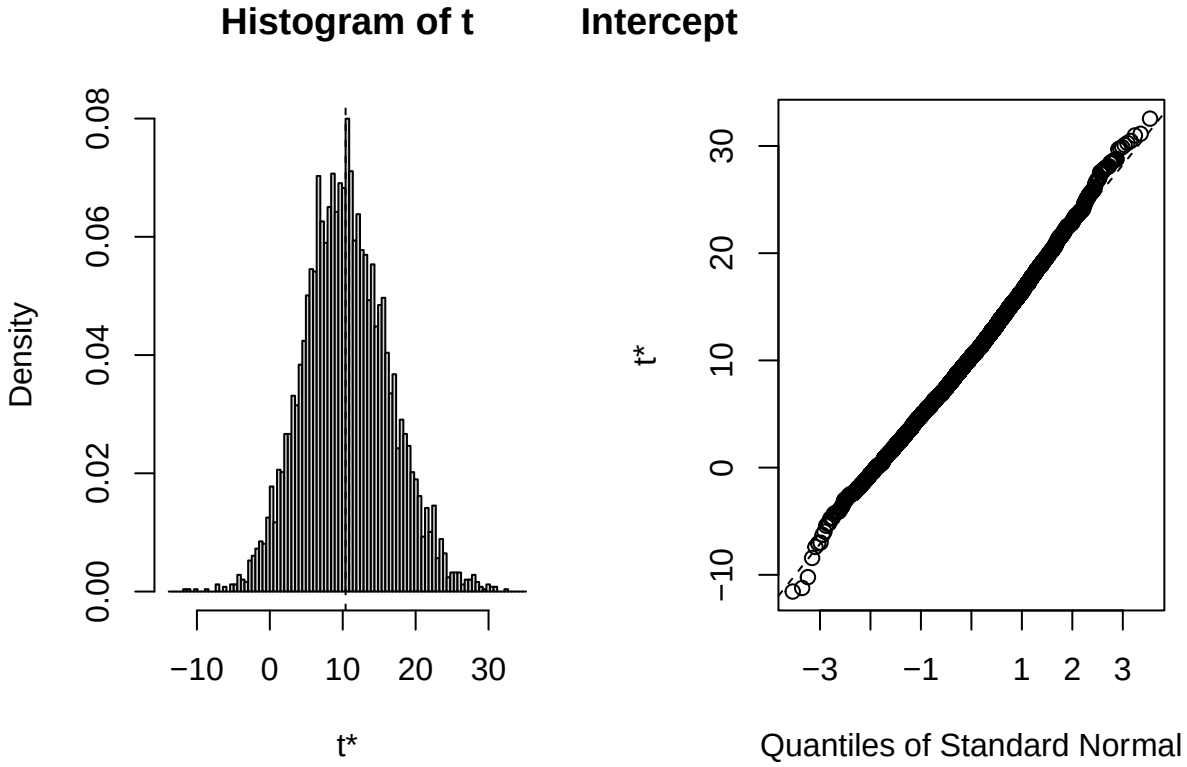
2.7.2) Confidence Intervals

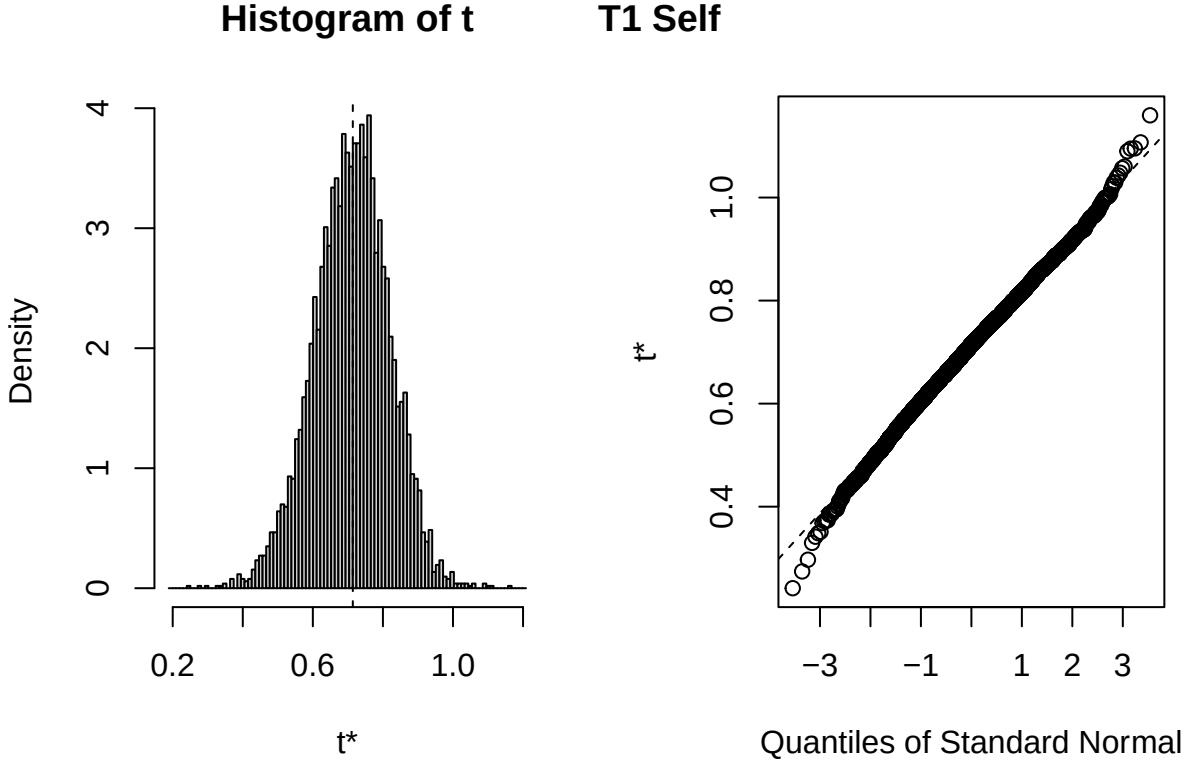
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb_t3, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.18, 22.92 )
## Calculations and Intervals on Original Scale

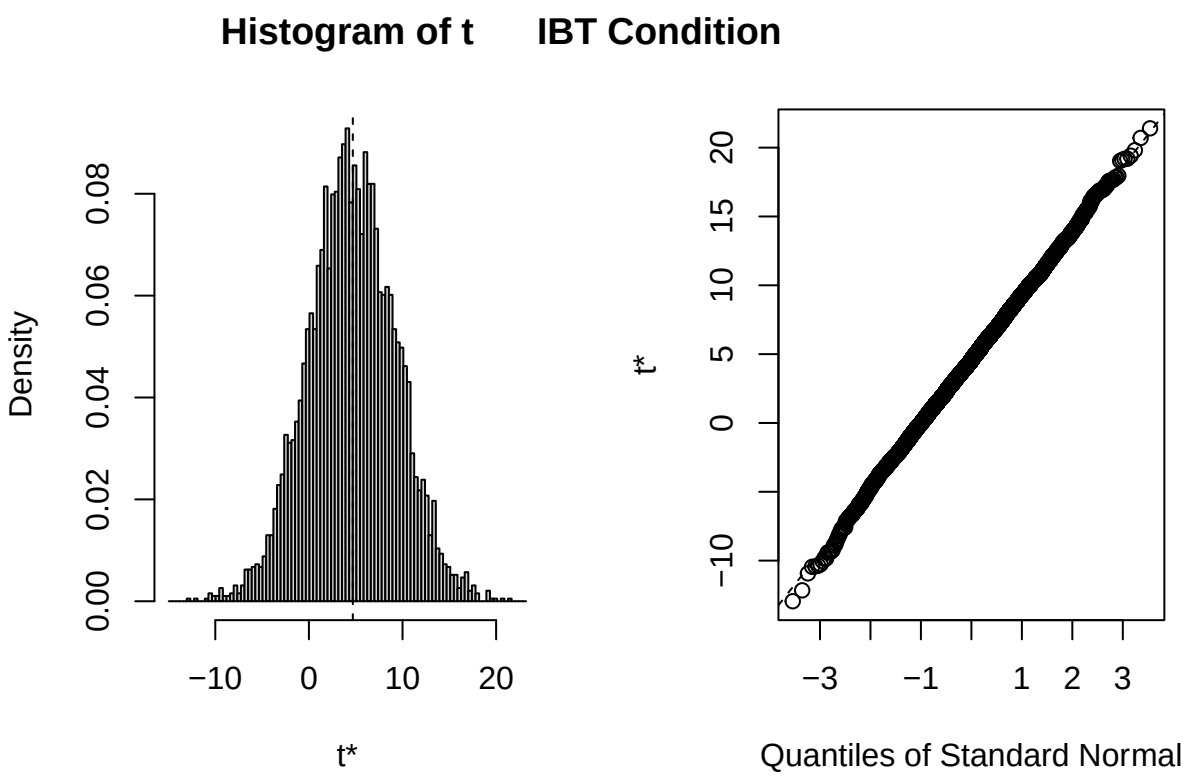
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb_t3, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4861,  0.9083 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb_t3, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-4.092, 14.076 )
## Calculations and Intervals on Original Scale
```

2.7.3) Bootstrap Diagnostics

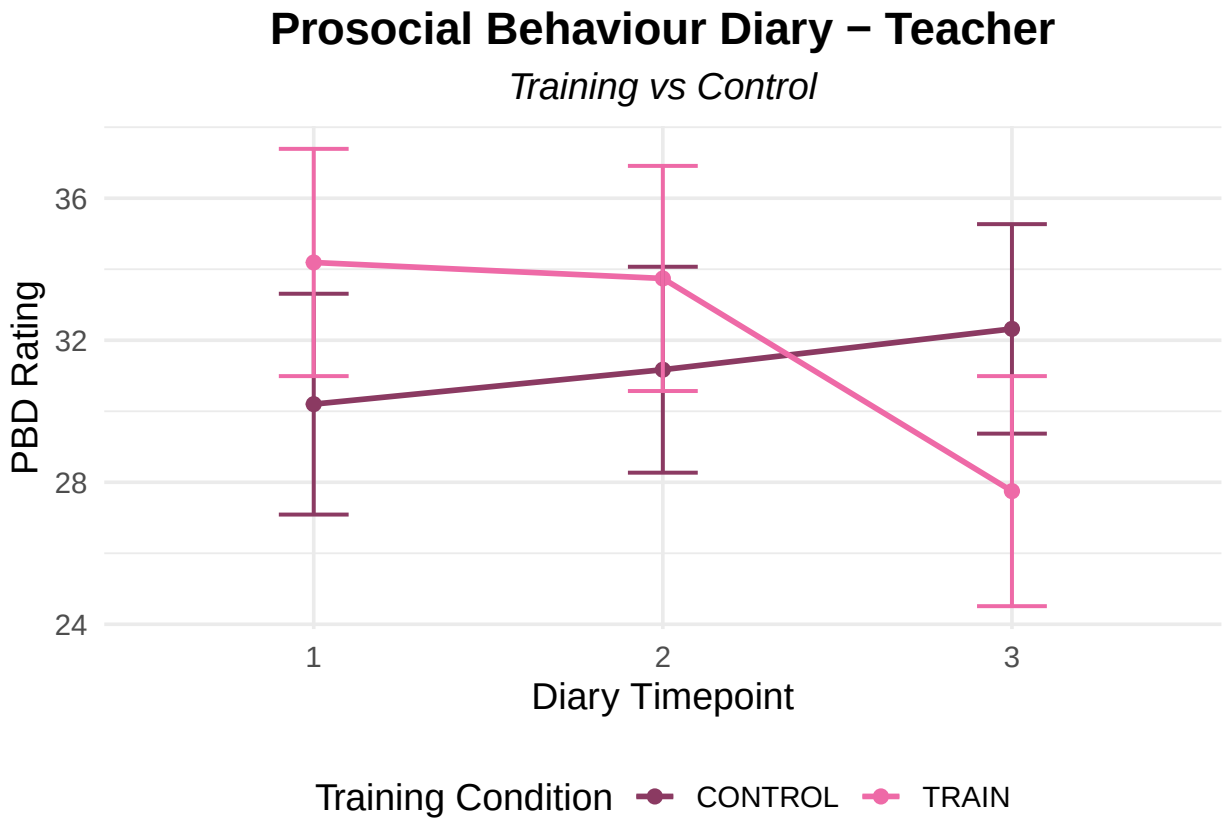




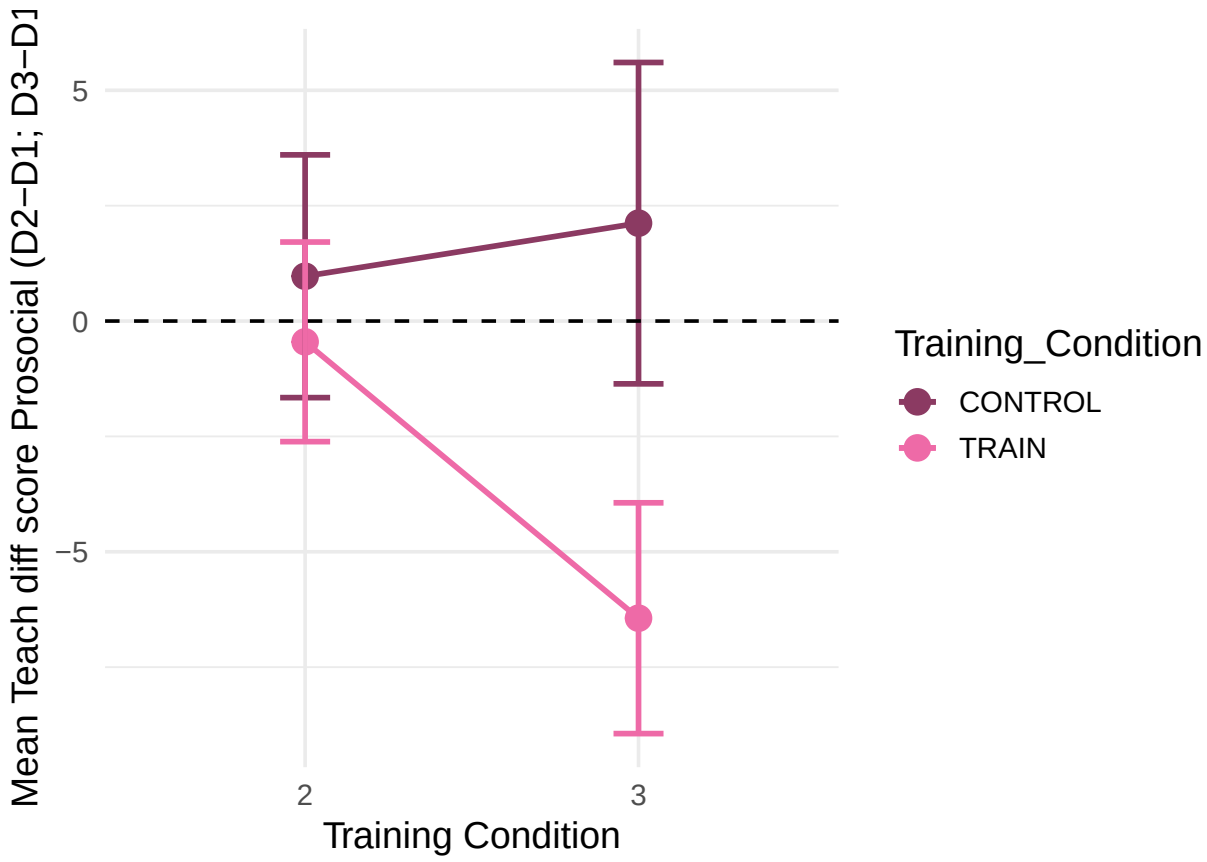

Histogram of t IBT Condition
Density
0.00
0.02
0.04
0.06
0.08
−10
0
10
20
t*
t*
−10
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal

3) Impact of training on teacher ratings of prosocial behaviour at two months (T3)

3.1) Plots of teacher prosocial ratings over time



Difference plots:



3.2) Preregistered LMM (wihtout random slopes)

```
# model that gives error:

# model_c_time_asb <- lmer(ABD_mean_rating ~ Mean_ABD_D1 + diaryTP +
#                            Training_Condition + diaryTP*Training_Condition +
#                        (1 + diaryTP|ID_numeric),
#                        data =  teach_abd_d2d3_longformat)

model_t_time_psb <- lmer(PBD_mean_rating ~ Mean_PBD_D1 + diaryTP +
                           Training_Condition + diaryTP*Training_Condition +
                      (1|ID_numeric),
                      data =  teach_pbd_d2d3_longformat)

summary(model_t_time_psb)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: PBD_mean_rating ~ Mean_PBD_D1 + diaryTP + Training_Condition +
##     diaryTP * Training_Condition + (1 | ID_numeric)
##    Data: teach_pbd_d2d3_longformat
##
## REML criterion at convergence: 1571.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.95934 -0.46586 -0.06022  0.39120  2.50326
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  ID_numeric (Intercept) 146.8    12.11
##  Residual               142.2    11.92
## Number of obs: 190, groups:  ID_numeric, 95
##
## Fixed effects:
##                                  Estimate Std. Error        df t value
## (Intercept)                       13.80551    3.18872 120.53933   4.329
## Mean_PBD_D1                        0.57499    0.07027  92.00000   8.183
## diaryTP3                           1.15033    2.36111  93.00000   0.487
## Training_ConditionTRAIN            0.27445    3.50855 146.03857   0.078
## diaryTP3:Training_ConditionTRAIN  -7.13896    3.46938  93.00000  -2.058
##                                  Pr(>|t|)
## (Intercept)                      3.11e-05 ***
## Mean_PBD_D1                      1.50e-12 ***
## diaryTP3                           0.6273
## Training_ConditionTRAIN            0.9378
## diaryTP3:Training_ConditionTRAIN   0.0424 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) M_PBD_ dryTP3 T_CTRA
## Mean_PBD_D1 -0.665
```

```
## diaryTP3    -0.370  0.000
## Trnn_CTRAIN -0.453 -0.080  0.336
## dTP3:T_CTRA  0.252  0.000 -0.681 -0.494
```

```
anova(model_t_time_psb)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                            Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## Mean_PBD_D1                9519.1  9519.1     1    92 66.9616 1.497e-12 ***
## diaryTP                     276.5   276.5     1    93  1.9448   0.16647
## Training_Condition          165.9   165.9     1    92  1.1673   0.28277
## diaryTP:Training_Condition  601.9   601.9     1    93  4.2342   0.04242 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
confint(model_t_time_psb)
```

```
## Computing profile confidence intervals ...
```

```
##                                     2.5 %     97.5 %
## .sig01                          9.2742519 14.7234835
## .sigma                         10.2992734 13.6958554
## (Intercept)                     7.6027153 20.0082974
## Mean_PBD_D1                     0.4380836  0.7119035
## diaryTP3                       -3.4750642  5.7757178
## Training_ConditionTRAIN        -6.5451802  7.0940808
## diaryTP3:Training_ConditionTRAIN -13.9354412 -0.3424852
```

```
confint(model_t_time_psb, level = 0.995)
```

```
## Computing profile confidence intervals ...
```

```
##                                    0.25 %    99.75 %
## .sig01                          8.0661709 16.1043635
## .sigma                          9.7485791 14.6770946
## (Intercept)                     4.8482219 22.7627908
## Mean_PBD_D1                     0.3767971  0.7731899
## diaryTP3                       -5.5455768  7.8462303
## Training_ConditionTRAIN        -9.5592588 10.1081593
## diaryTP3:Training_ConditionTRAIN -16.9778203  2.6998939
```

3.2.1) Cmplexity check: does interaction improve model? - It does

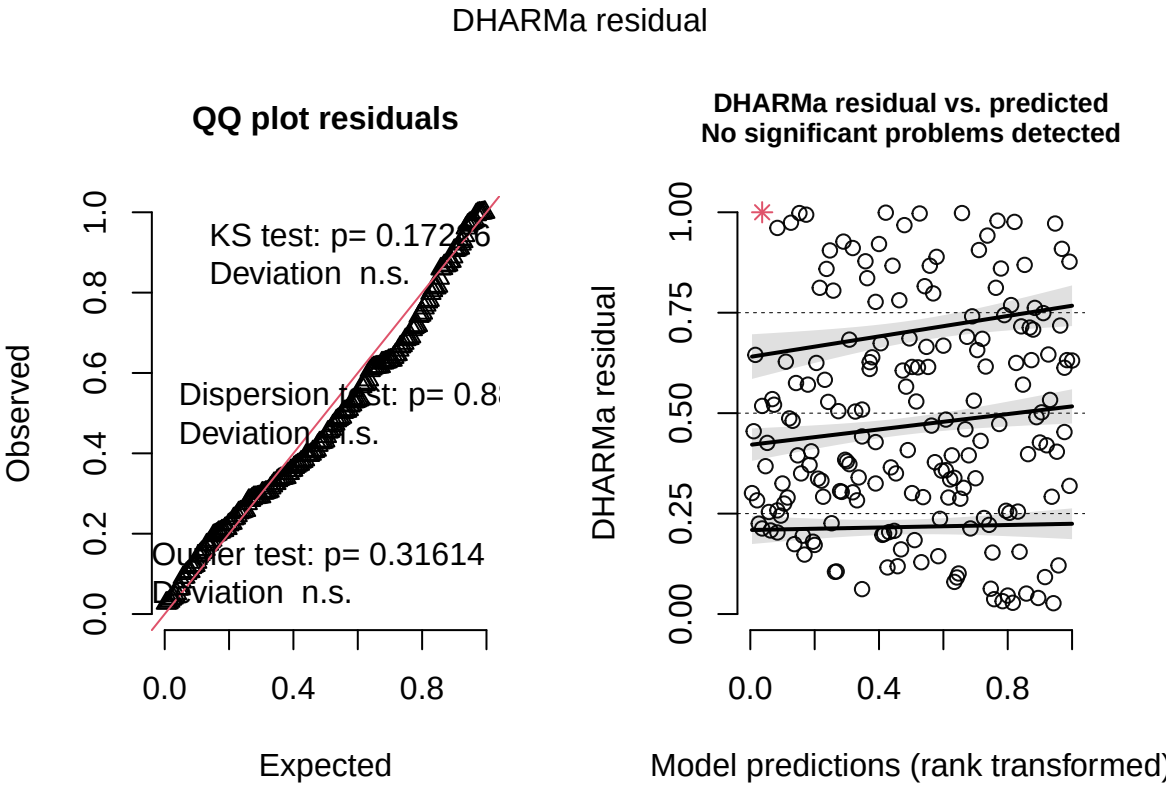
```
model_t_time_psb_simple <- lmer(PBD_mean_rating ~ Mean_PBD_D1 + diaryTP +
                          Training_Condition +
                      (1|ID_numeric),
                      data =  teach_pbd_d2d3_longformat)

anova(model_t_time_psb, model_t_time_psb_simple)
```

```
## refitting model(s) with ML (instead of REML)

## Data: teach_pbd_d2d3_longformat
## Models:
## model_t_time_psb_simple: PBD_mean_rating ~ Mean_PBD_D1 + diaryTP + Training_Condition + (1 | ID_nume
## model_t_time_psb: PBD_mean_rating ~ Mean_PBD_D1 + diaryTP + Training_Condition + diaryTP * Training_
##                          npar    AIC    BIC  logLik -2*log(L)  Chisq Df
## model_t_time_psb_simple    6 1598.6 1618.0 -793.29    1586.6
## model_t_time_psb           7 1596.3 1619.1 -791.17    1582.3 4.2297  1
##                          Pr(>Chisq)
## model_t_time_psb_simple
## model_t_time_psb           0.03972 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.2.2) Assumption check



3.2.3) Bootstrapped prereg LMM

```
set.seed(123)  # For reproducibility
boot_results <- lmeresampler::bootstrap(model_t_time_psb, .f = fixef, type = "reb", B = 1000)
```

3.2.4) Bootstrap results and CIs

```
## Bootstrap type: reb0
##
## Number of resamples: 1000
##
##                                term   observed   rep.mean         se
## 1                       (Intercept) 13.8055064 13.8940389 3.26134940
## 2                       Mean_PBD_D1  0.5749935  0.5708092 0.07334271
## 3                          diaryTP3  1.1503268  1.1657127 1.62132484
## 4           Training_ConditionTRAIN  0.2744503  0.4461704 3.51729309
## 5 diaryTP3:Training_ConditionTRAIN -7.1389632 -7.1829787 2.55069583
##           bias
## 1  0.088532504
## 2 -0.004184332
## 3  0.015385947
## 4  0.171720147
## 5 -0.044015580
##
## There were 0 messages, 0 warnings, and 0 errors.
```

```
## # A tibble: 5 x 6
##   term                              estimate   lower  upper type  level
##   <chr>                                <dbl>   <dbl>  <dbl> <chr> <dbl>
## 1 (Intercept)                          13.8     5.60  22.8   perc   0.99
## 2 Mean_PBD_D1                           0.575   0.395  0.755 perc   0.99
## 3 diaryTP3                              1.15   -2.96   5.38  perc   0.99
## 4 Training_ConditionTRAIN               0.274  -8.24   9.19  perc   0.99
## 5 diaryTP3:Training_ConditionTRAIN     -7.14  -13.2   -0.799 perc   0.99
```

3.3) UPDATED FROM PREREG: LMM predicting T3, adjusted for T1 - singular fit error

```
model_t_time_psb_d3only_full <- lmer(PBD_mean_rating ~ Mean_PBD_D1 +
                          Training_Condition +
                        (1|School.ID/teachUUID_diaries),
                        data =  teach_pbd_d3_only)
```

```
## boundary (singular) fit: see help('isSingular')
```

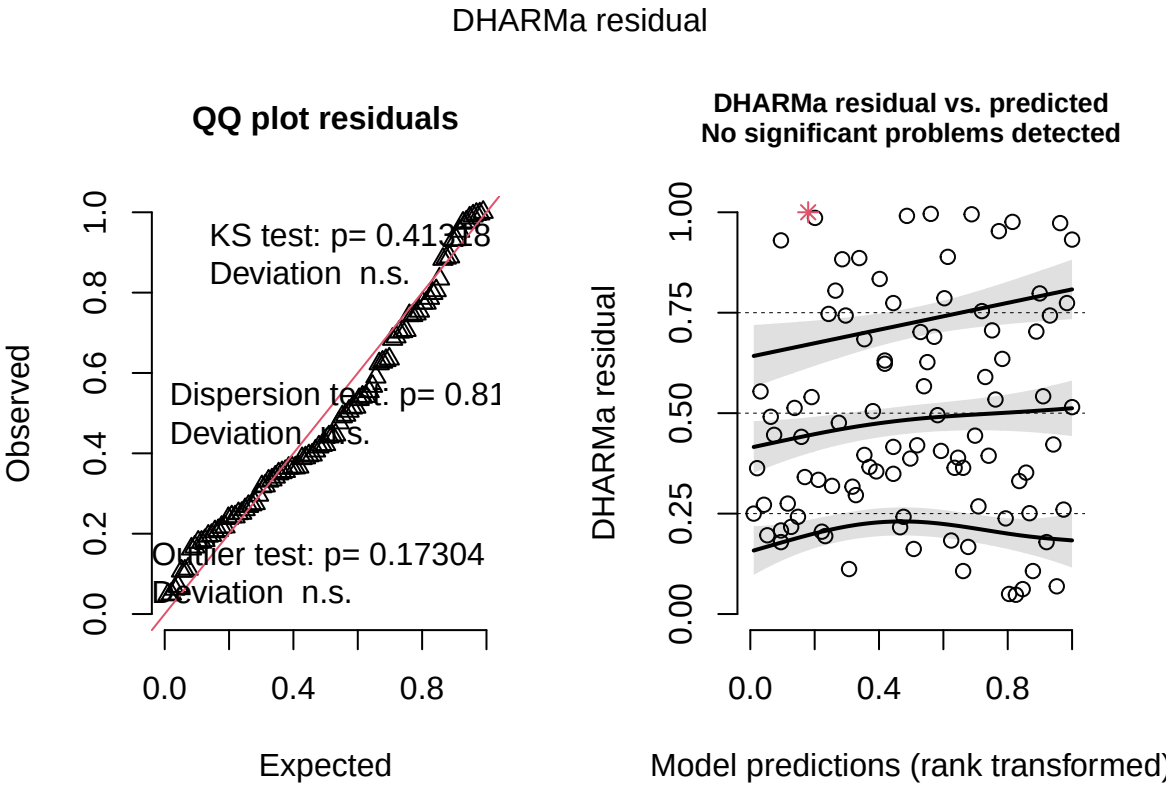
3.4) LMM predicting T3, adjusted for T1 - with only teach random effect

```
model_t_time_psb_d3only <- lmer(PBD_mean_rating ~ Mean_PBD_D1 +
                          Training_Condition +
                        (1|teachUUID_diaries),
                        data =  teach_pbd_d3_only)


summary(model_t_time_psb_d3only)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition + (1 | teachUUID_diaries)
##    Data: teach_pbd_d3_only
##
## REML criterion at convergence: 811.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.1340 -0.6038 -0.1226  0.3799  2.9661
##
## Random effects:
##  Groups            Name        Variance Std.Dev.
##  teachUUID_diaries (Intercept)  90.42    9.509
##  Residual                      262.81   16.211
## Number of obs: 95, groups:  teachUUID_diaries, 35
##
## Fixed effects:
##                           Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)              17.57239    3.86810 77.50942   4.543 2.01e-05 ***
## Mean_PBD_D1               0.44852    0.09257 88.76685   4.845 5.33e-06 ***
## Training_ConditionTRAIN -5.48822    3.68952 88.32466  -1.488     0.14
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) M_PBD_
## Mean_PBD_D1 -0.640
## Trnn_CTRAIN -0.321 -0.163
```

3.4.1) Assumption Check:



3.5) Bootstrapped LMM - this gave singular fit on one iteration. Decided to simplify in doubt/for consistency.

```
set.seed(1423)  # For reproducibility
boot_results <- lmeresampler::bootstrap(model_t_time_psb_d3only, .f = fixef, type = "reb", B = 5000)


boot_results
```

```
## Bootstrap type: reb0
##
## Number of resamples: 5000
##
##                       term   observed   rep.mean         se         bias
## 1              (Intercept) 17.5723856 16.5088577 3.90805966 -1.063527810
## 2              Mean_PBD_D1  0.4485242  0.4465215 0.08519679 -0.002002688
## 3 Training_ConditionTRAIN -5.4882181 -5.4790886 3.15690308  0.009129471
##
## There were 1 messages, 0 warnings, and 0 errors.
##
## The most commonly occurring message was: boundary (singular) fit: see help('isSingular')
```

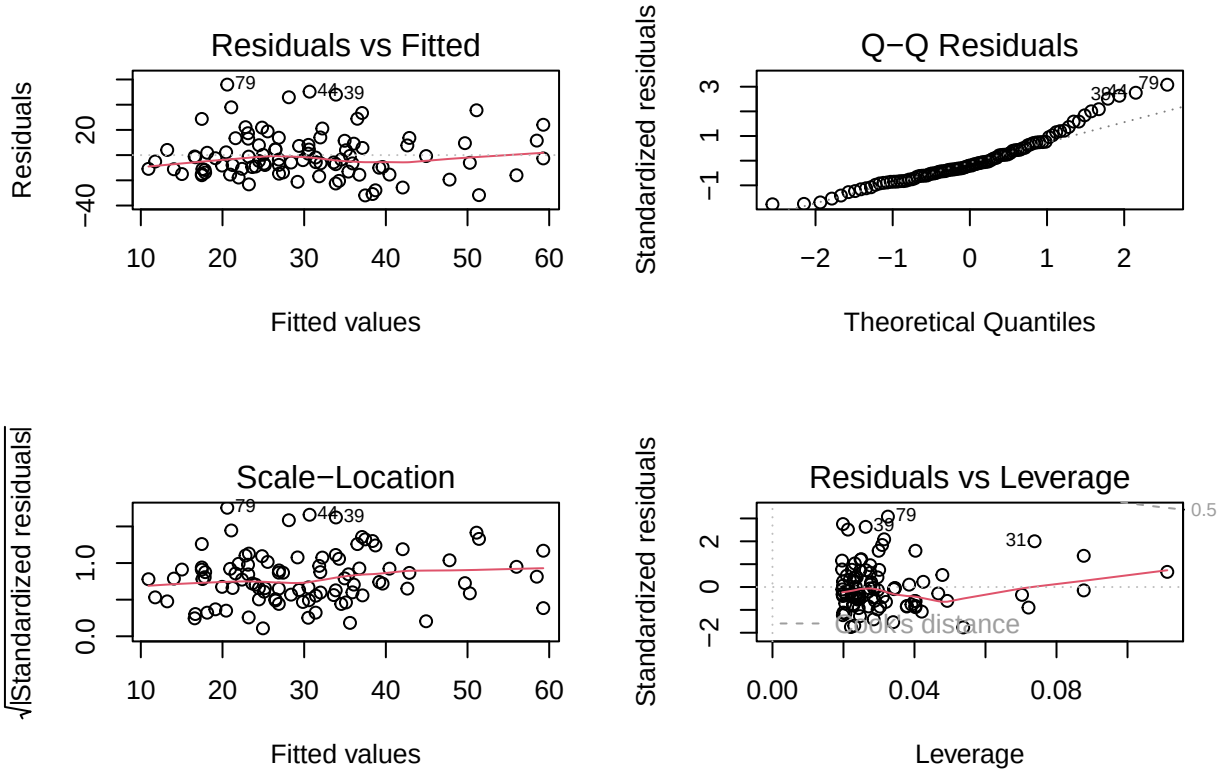
3.6) Linear Regression

```
model_t_time_psb_d3only_lm <- lm(PBD_mean_rating ~ Mean_PBD_D1 +
                           Training_Condition,
                       data =  teach_pbd_d3_only)

summary(model_t_time_psb_d3only_lm)
```

```
##
## Call:
## lm(formula = PBD_mean_rating ~ Mean_PBD_D1 + Training_Condition,
##     data = teach_pbd_d3_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -31.969 -11.518  -3.560   8.616  55.918
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              17.46670    3.70508   4.714 8.60e-06 ***
## Mean_PBD_D1               0.49185    0.08789   5.596 2.24e-07 ***
## Training_ConditionTRAIN -6.53277    3.81463  -1.713   0.0902 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.46 on 92 degrees of freedom
## Multiple R-squared:  0.2626, Adjusted R-squared:  0.2466
## F-statistic: 16.38 on 2 and 92 DF,  p-value: 8.204e-07
```

3.6.1) Assumption Checks:



3.7) Final Model: Bootstrapped Linear Regression

```
boot_coef_t_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PBD_mean_rating ~ Mean_PBD_D1 +
                          Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
boot_results_t_psb <- boot(data = teach_pbd_d3_only,
                     statistic = boot_coef_t_psb,
                     R = 5000)
```

3.7.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_pbd_d3_only, statistic = boot_coef_t_psb, R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 17.4666959  0.179755205   4.1316573
## t2*  0.4918506 -0.003498867   0.0921573
## t3* -6.5327686 -0.074726244   3.6377000
```

3.7.2) Confidence Intervals - 95%

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (10.03, 26.35 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3008,  0.6572 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-13.693,   0.632 )
## Calculations and Intervals on Original Scale
```

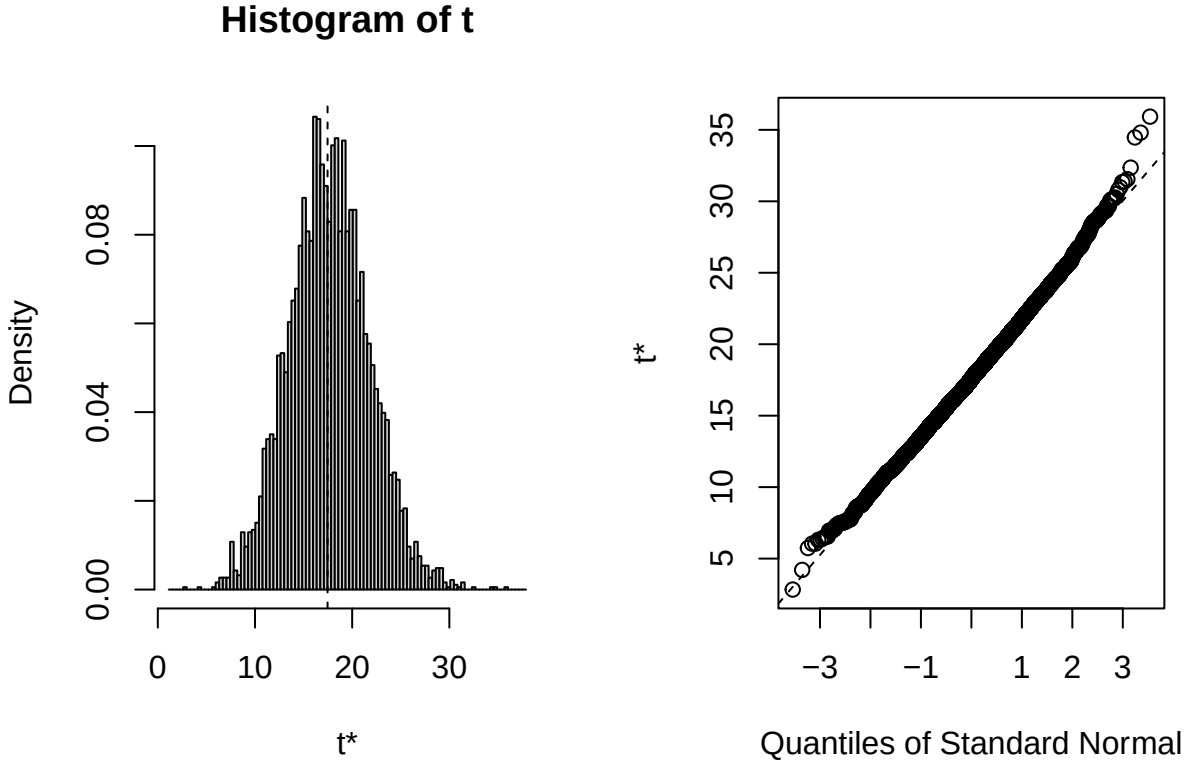
3.7.3) Confidence Intervals - 99%

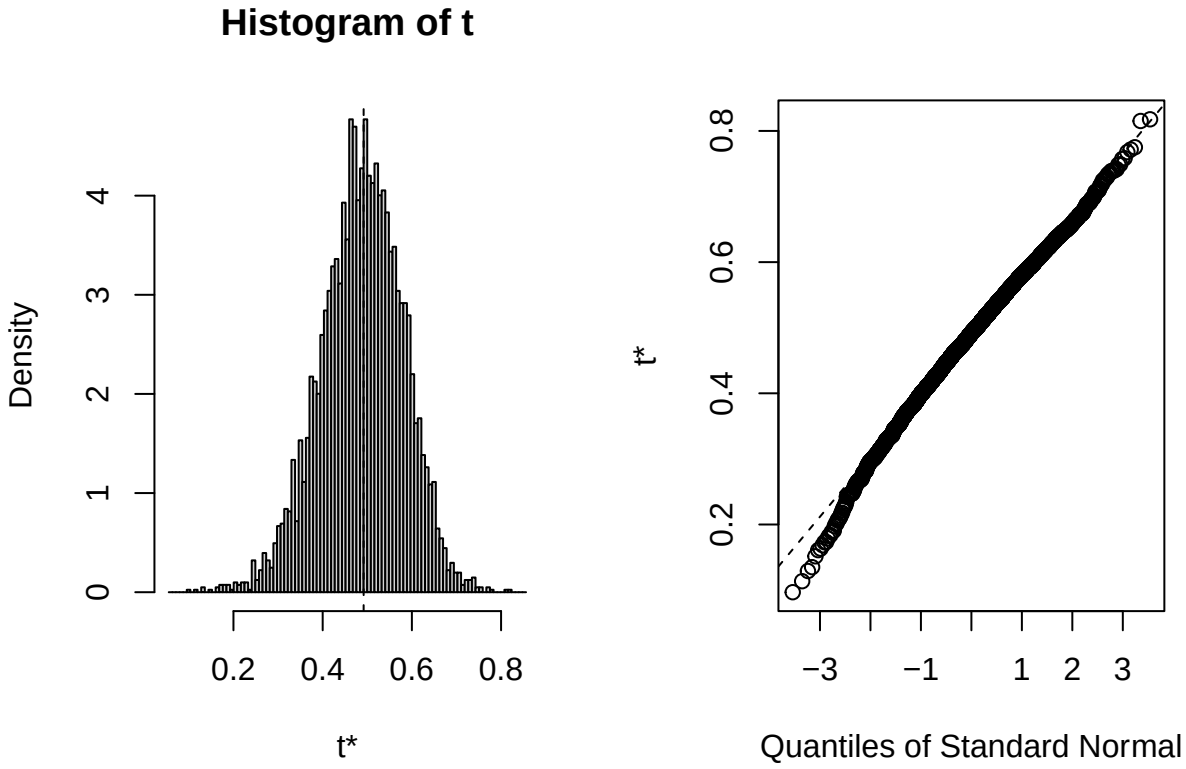
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, conf = 0.99, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 99%   ( 7.74, 29.94 )
## Calculations and Intervals on Original Scale

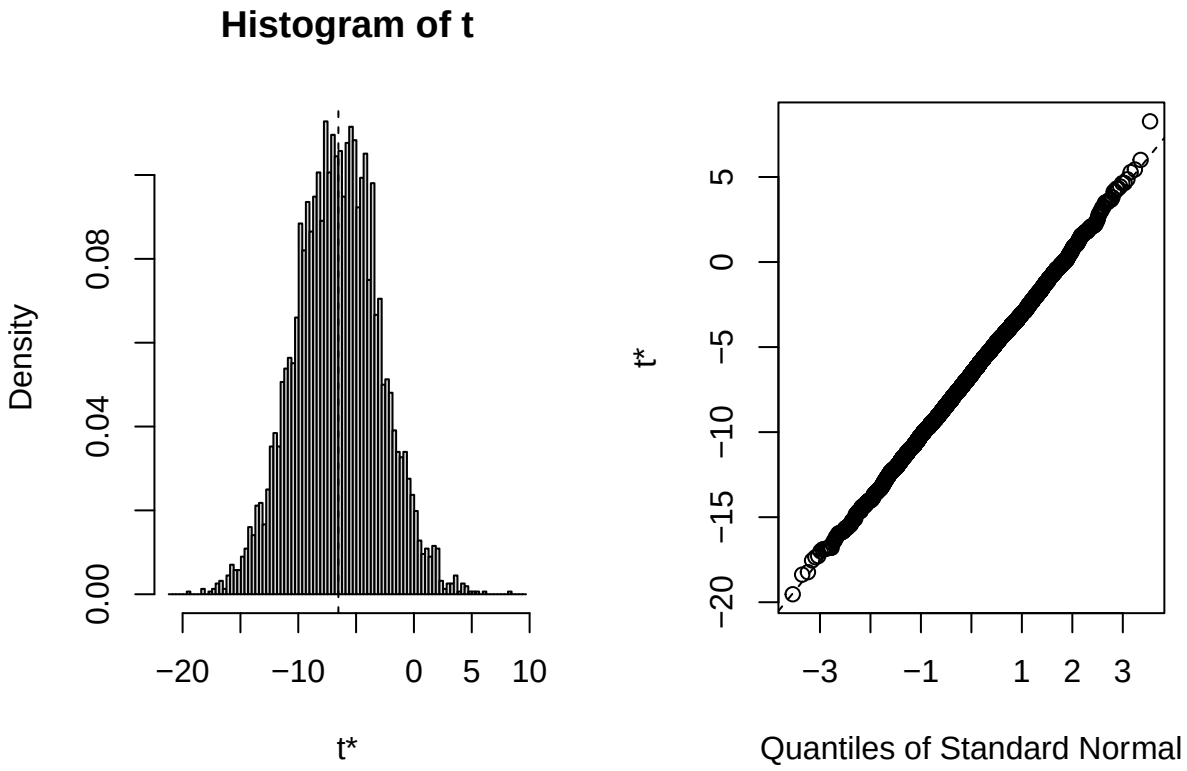
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, conf = 0.99, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 99%   ( 0.2231,  0.7246 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, conf = 0.99, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 99%   (-15.854,   3.193 )
## Calculations and Intervals on Original Scale
```

3.7.4) Bootstrap Diagnostics






Histogram of t
Density
0.00
0.04
0.08
−20
−10
0
5
10
t*
t*
−20
−15
−10
−5
0
5
−3
−1
1
2
3
Quantiles of Standard Normal